Differential splicing between cell types from split-pool single-cell RNA-seq with random-hexamer priming

Abstract

Combinatorial-indexing single-cell RNA-seq profiles hundreds of thousands of cells from tens of donors in one experiment, and its in-cell reverse transcription with random hexamers alongside poly(dT) covers transcript bodies, so it observes the splice junctions of most expressed genes in every cell type of every donor. Existing methods for differential splicing were designed for bulk or full-length data and use only junction-spanning reads summed over cells. We present Tealeaf, which tests local splice paths, the exon chains that transcripts take through a region bounded by constitutive exons, so that each result names the transcript structures that change, such as a cassette exon or an alternative first exon, rather than a shift among junctions. Tealeaf uses every molecule that overlaps the region through the set of paths it is compatible with, models poly(dT) and random-hexamer molecules in one likelihood with a primer offset estimated without cell-type labels, builds unannotated paths from junction support counted over all cells, and treats donors as replicates with a Beta–logit-normal model with a donor random effect. In a mouse brain dataset of 48 animals, Tealeaf replicates more genes across independent halves of the animals than LeafCutter, MAJIQ, scQuint, rMATS and two forms of SUPPA2, its 100 top-ranked tests agree in direction with single-cell long reads as often as the best comparator's, and it has 2.5 to 26 times as many tests that the long reads can evaluate. A coverage-only null that we apply to every method shows that MAJIQ and rMATS call many tests from expression differences between cell types alone. In three human blood datasets, Tealeaf's top-ranked tests agree with single-cell long reads of a Parse library and its tests between groups of different donors are calibrated, while LeafCutter replicates more genes than Tealeaf in the largest dataset.

1 Introduction

Alternative splicing differs between tissues and cell types [1], most of all in the nervous system [2], and single-cell RNA-seq can in principle measure it in every cell type of a tissue at once [3]. Droplet methods read the $3'$ or $5'$ end of each transcript [4], so they observe few splice junctions. Plate methods read whole transcripts [5, 6] but profile few cells and donors.

Combinatorial-indexing methods barcode cells in successive rounds of split-pool reactions [7–11], so one experiment profiles hundreds of thousands of cells or nuclei from tens of samples, each sample in its own first-round wells. In SPLiT-seq and its commercial form (Parse Biosciences), the first round is reverse transcription inside the cell with a mixture of barcoded poly(dT) and random-hexamer primers [8], and EasySci combines the two primers in the same way [11]. Random-hexamer priming starts along the whole transcript, and the primer of every read is known from its barcode. These data therefore observe splice junctions across transcript bodies, in many cell types and in many donors, the design needed to test splicing differences with donors as biological replicates. In a Parse dataset of mouse brain nuclei from 48 animals that we analyse here (the brain data), random-hexamer molecules are 45% of the molecules that overlap alternatively spliced regions and 61% of those that span a junction distinguishing the alternatives.

Methods for differential splicing were not designed for these data. Event methods (SUPPA2 [12], rMATS [13]) test predefined event types, and junction methods (LeafCutter [14], MAJIQ [15, 16], scQuint [17]) test the relative use of junctions that share a splice site or an exon. A junction cluster whose usage changes does not identify the transcript structure that changes, since a cassette exon, an alternative first exon and two neighbouring alternative splice sites all appear as shifts among junctions that share a splice site, and the event has to be inferred afterwards. Both kinds of method use junction-spanning reads of pseudobulk samples, the reads of one donor and cell type summed over its cells. Single-cell methods model cells rather than donors [18, 19], and treating cells as replicates inflates false discoveries [20, 21]. None uses the reads inside exon bodies, which are most of the reads in a short-read library, and none models the two primers, which sample transcripts with different positional bias.

We present Tealeaf (Testing Exon Alternatives Locally in Exon-Anchored Frames), which tests whether the usage of local splice paths differs between two cell types measured in the same donors, or between two groups of donors within a cell type. A path is a whole exon chain through a region, annotated or not, so a Tealeaf result reads as an event of transcript structure with an effect on the scale of molecule shares. We evaluate Tealeaf against six comparators by replication across independent halves of the donors and by agreement of direction with independent single-cell long-read data, after removing, for every method, the tests that a coverage-only null also calls, a null in which each sample keeps its coverage and the two cell types share one usage.

2 Results

2.1 Overview of Tealeaf

Tealeaf goes from paired FASTQ files to tests in five steps (Fig. 1; Supplementary Fig. S1): targeted spliced alignment, blocks and paths, molecule classes, read opportunities and the test.

The unit of test is a *splice block*, a window of one gene bounded by constitutive exons in which transcripts take two or more exon chains, the *paths* (Fig. 1a). A path set represents any local pattern, including several adjacent alternative exons, mutually exclusive exons, combinations of alternative splice sites and alternative first or last exons, in one test, and the paths themselves show which exons and splice sites differ. We take the paths of annotated blocks from the annotation, and we add novel paths (alternative splice sites, new exons and novel first and last exons) and new blocks from unannotated junctions, whose read support we count over all cells and therefore without cell-type labels (Supplementary Fig. S2). In the brain data the block set has 34,677 blocks, 11,779 of them new, with 21,509 novel paths. We call a block *internal* if all its annotated paths share their first and last exon, and *terminal* if they differ at a transcript end (alternative first or last exons, or transcript starts or ends). For reporting, we assign each test an *event class*, such as cassette exon or alternative first exon, from the exon chains of its tested paths (Methods).

We group reads into molecules by cell barcode and unique molecular identifier (UMI), and assign each molecule that overlaps a block its *class*, the set of paths it is compatible with, from the parts of its reads inside the block window. A read inside an alternative exon is compatible with the paths that include the exon, and a read inside a constitutive exon with all paths (Fig. 1a). For every path and class, the *read opportunities*, the number of read start positions on the path that produce the class, give the class probabilities as a function of path usage, so exon-body and junction reads enter one likelihood in proportion to the information they carry. In terminal blocks, exon-body coverage follows the positional capture of the primers (Fig. 1b), so for these blocks we use only molecules that span a junction distinguishing the paths (the *junction-only* test). Poly(dT) read ends are also spread along transcript bodies, at about the same density per kilobase near the $5'$ end as in

the middle, apart from the last 200 bases (Fig. 1b), because poly(dT) primes at A-rich sequence inside transcripts (Methods). Since this capture depends on the sequence of each path, poly(dT) and random-hexamer molecules of the same row can imply different usage, which the primer offset absorbs.

A test compares one block between two cell types (Fig. 1c). We test each path against the others with a Beta–logit-normal model in which every donor–cell-type row has its own usage, rows vary around a mean that depends on the cell type through a donor random intercept, and the poly(dT) and random-hexamer molecules of a row share the row's usage up to a primer offset that we estimate without cell-type labels. A likelihood-ratio test of the cell-type effect gives one p-value per path, and we combine the path p-values by Simes' method [22]. Because the donor intercept is shared by the donor's two cell types, the test is paired. The same model compares two groups of different donors within a cell type, with one row per donor. In count simulations under the null on the designs of the brain data, internal tests reject at most 2.1% of converged fits at nominal 0.05 and at most 0.7% at 0.01, and junction-only terminal tests 2.0 to 4.0% at 0.05.

We align reads to the loci of the target genes with a two-pass spliced aligner that reproduces the scoring and filters of STAR [23] and replaces the whole-genome search for competing alignments by a genome-wide 23-mer uniqueness table (Methods). Compared with STARsolo [24], the single-cell mode of STAR, it reports 97.6% of STAR's unique alignments in the target loci, with identical junctions for 99.5% of reads that both align, and molecule counts per block, class, donor, cell type and primer agree to within 5%. A run of 345 million read pairs aligns to all 33,755 multi-exon genes in 32 to 67 minutes, against 67 minutes for STARsolo over the whole genome, and its stored alignments take about 1 GB instead of a 46 GB BAM file.

2.2 Evaluation design

We evaluated Tealeaf on the brain data, nuclei of 48 animals in six groups of eight (wild type, the microglia-deficient FIRE line [25], 5xFAD [26], 5xFAD crossed to FIRE [27], and the cross treated with PBS or with a microglia transplant), with 37,002 nuclei in 20 cell types and 5.6 billion read pairs. We test every pair of cell types, and call each pair a contrast.

We use two endpoints, neither of which needs knowledge of the true splicing differences. For *split replication*, we divide the animals into two halves of 24, balanced by group, and run each method on each half. A gene replicates when it has signal in both halves, by the Benjamini–Hochberg (BH) [28] adjustment over genes of the larger of its two half p-values, each the Simes combination of the gene's tests in that half. Each method tests its own (gene, cell-type pair) keys, so we score each comparison on two sets of keys, the comparator's *universe*, the keys it tests in both halves, and the *matched universe*, the subset that Tealeaf also tests in both halves. For *long-read direction*, we compare the effects of the tests on the full data (all 48 animals) with the P56 mouse brain single-cell long-read data of Joglekar et al. [29], in eight cell types with a prespecified correspondence. We assign long-read transcripts to the tested paths (or junctions or events) of every method by their exon chains in the comprehensive annotation, and call a test *evaluable* when the long reads cover its tested paths in both cell types (Methods). We rank the evaluable tests by p-value, and the endpoint A_{100} is the mean, over $r = 1, \ldots, 100$, of the fraction of the r top-ranked tests whose direction agrees with the long reads. Split replication does not use direction, and long-read direction does not use replication.

The comparators are LeafCutter, MAJIQ Heterogen, scQuint, paired rMATS (PAIRADISE [30]), SUPPA2 with paired signed-rank tests, and a primer-aware SUPPA2 that estimates event usage from equivalence-class counts of each primer with a primer offset (Methods). We run each on the same donor–cell-type pseudobulks, contrasts and halves. For every method we remove, before

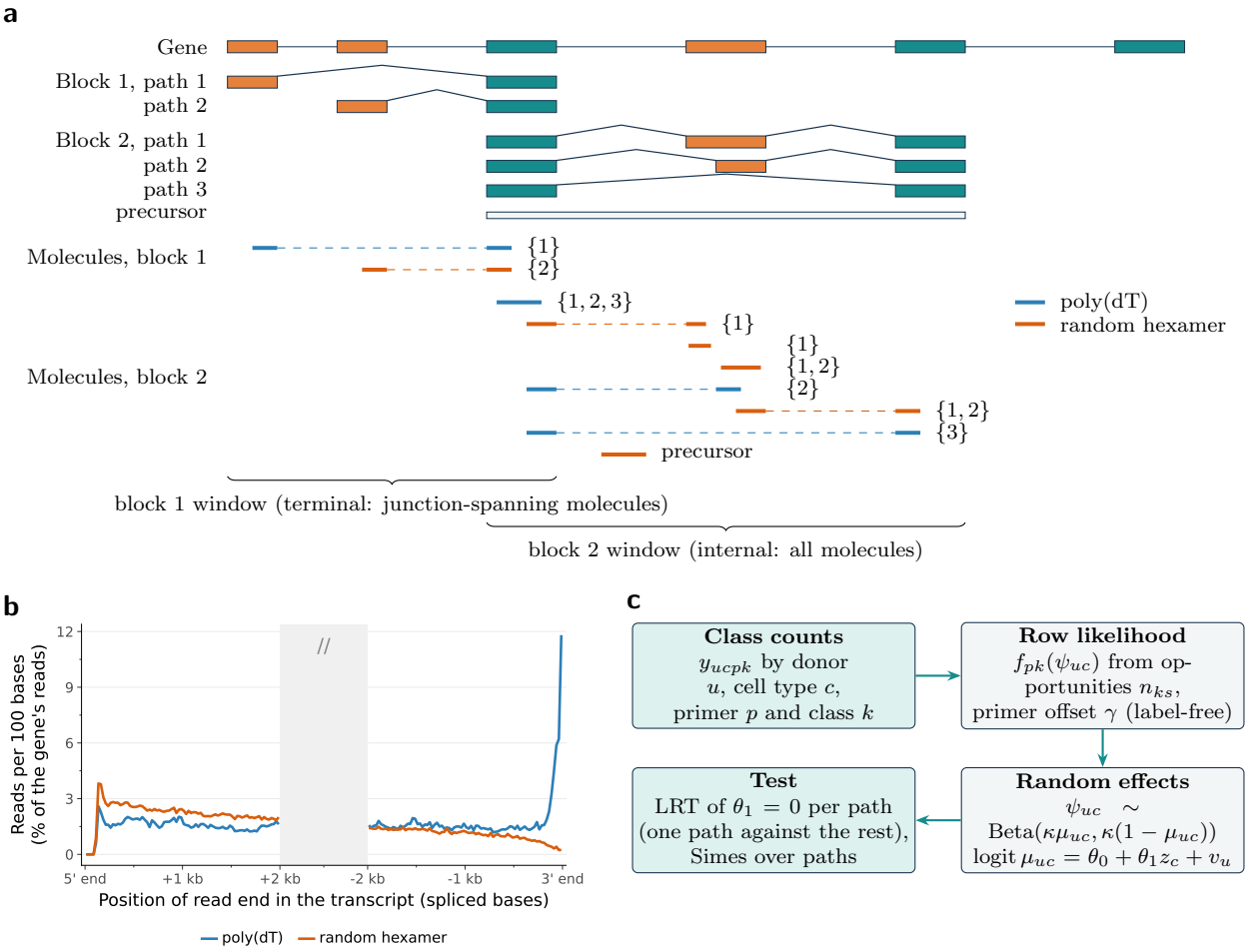


Figure 1: Tealeaf. **a**, A gene with two splice blocks between constitutive exons (teal), and a constitutive last exon in no block. Block 1 is terminal, with two alternative first exons (orange); block 2 is internal, with an alternative exon included with an upstream (path 1) or downstream (path 2) 3′ splice site or skipped (path 3), and a precursor path (the unspliced window). Below are molecules and their classes, the set of paths compatible with a molecule's reads inside the block window: a read in the part of the exon used only by path 1 is compatible with path 1 alone, a read in the shared part with paths 1 and 2, and a read in an anchor exon with all three. In terminal blocks only molecules that span a distinguishing junction are used. The read opportunities n_{ks} count the read start positions on path s that give class k. **b**, Read ends along transcripts for poly(dT) and random-hexamer molecules of the brain data: the mean, over 1,500 to 1,700 long single-isoform genes per library, of each gene's share of its reads per 100 bases within 2 kb of the 5′ end (left) and of the 3′ end (right, oriented 5′ to 3′); the middle of the transcripts is omitted. **c**, The test. Class counts of each donor u, cell type c and primer p follow from the row usage ψ_{uc} through the opportunities, with a primer offset γ; ψ_{uc} varies around a mean with a cell-type effect θ_1 and a donor intercept v_u, each path is tested against the others by a likelihood-ratio test (LRT) of $\theta_1 = 0$, and the path p-values are combined by Simes' method.

both endpoints, the tests that the coverage-only null also calls (Methods). In this null each sample keeps its coverage of the tested feature and the two cell types share one usage, so a test it calls responds to coverage, not to splicing. We recompute each test with $p < 0.01$ on the data on three null draws and remove it when the median of its three null p-values is below 0.05, which removes about 0.7% of the recomputed tests of a calibrated method. The filter removes 0.4 to 0.9% of the recomputed tests of Tealeaf, LeafCutter and scQuint, 8 to 12% for rMATS and 43 to 68% for MAJIQ (Supplementary Table S7). We do not recompute SUPPA2, which replicates 0 to 8 genes.

2.3 Tealeaf replicates more genes in mouse brain

On each comparator's matched universe Tealeaf replicates more genes than the comparator (Fig. 2a): 182 against 165 for LeafCutter, 205 against 17 for MAJIQ, 192 against 134 for scQuint, 167 against 4 for rMATS, 130 against 8 for SUPPA2 and 138 against 0 for primer-aware SUPPA2. The comparator universes give the same order (Supplementary Table S1). The matched universes differ in size from 1,509 genes for LeafCutter to 5,070 for rMATS, because each method tests the genes where its own features have data in both halves.

Tealeaf's 100 top-ranked long-read-evaluable tests agree with the long reads in direction for 95 ($A_{100} = 0.947$), MAJIQ's after the filter for 90 (0.944), and those of the other comparators for 69 to 90 (Fig. 2b). Agreement holds further down the ranking. At rank 200 Tealeaf agrees for 94% of its tests against 84 to 86% for the junction methods and 81% for rMATS. Tealeaf has 35,218 evaluable tests, against 1,351 to 13,989 features for the junction methods and 2,773 to 4,491 events for rMATS and SUPPA2 on their 500 top-ranked events per contrast (Fig. 2c).

On the 28 pairs of the long-read cell types, the contrasts every method tests, Tealeaf makes 3,633 discoveries (BH-adjusted $p \leq 0.05$ within method) in 3,028 gene–cell-type pairs, scQuint 2,205 pairs, LeafCutter 1,634, SUPPA2 1,221, primer-aware SUPPA2 412, MAJIQ 263 and rMATS 164 (Fig. 2d). Tealeaf discovers 1,022 cassette exons, SUPPA2 1,123 and rMATS 166, and 1,480 of Tealeaf's discoveries are alternative first or last exons, which the brain event sets of SUPPA2 and rMATS do not contain. Over all 190 pairs, Tealeaf has 9,013 gene–cell-type pairs with a discovery, scQuint 8,742, LeafCutter 4,717 and SUPPA2 3,658 (Supplementary Fig. S3). By event class, long-read agreement of Tealeaf's evaluable discoveries is 0.83 to 0.90 for the common classes, and 6,994 of its 10,830 full-data discoveries retain an unannotated path (Supplementary Table S2).

2.4 Coverage differences produce calls of MAJIQ and rMATS

Without the coverage-only filter, MAJIQ and rMATS replicate 27 and 4 genes on their matched brain universes, and 991 and 184 on their universes in the human blood dataset GSE320507 described below. To see whether these replications reflect splicing, we recomputed each method's tests on data with the observed coverage and no difference in usage between the cell types. MAJIQ Heterogen compares per-sample estimates of percent spliced in (PSI) whose prior pulls them toward a fixed value at low coverage, so a gene expressed at different levels in two cell types gives significant tests without a difference in PSI (Supplementary Note 2). Under the coverage-only null its call rate follows the ratio of the feature's coverage between the two cell types and reaches 93% of tests when the ratio exceeds eightfold, against 94% on the data (Fig. 3a). Because expression differences between cell types are shared by all donors, these calls replicate across halves. In the same null, Tealeaf rejects 0 to 4% of internal and 2 to 5% of terminal tests at nominal 0.05 in every class of the coverage ratio, and LeafCutter and scQuint make no discoveries. rMATS, run through PAIRADISE on four whole contrasts of null counts, makes about as many BH calls of mutually exclusive exons as on the data and more of skipped exons (Fig. 3c), rejecting 10 to 13% of skipped-

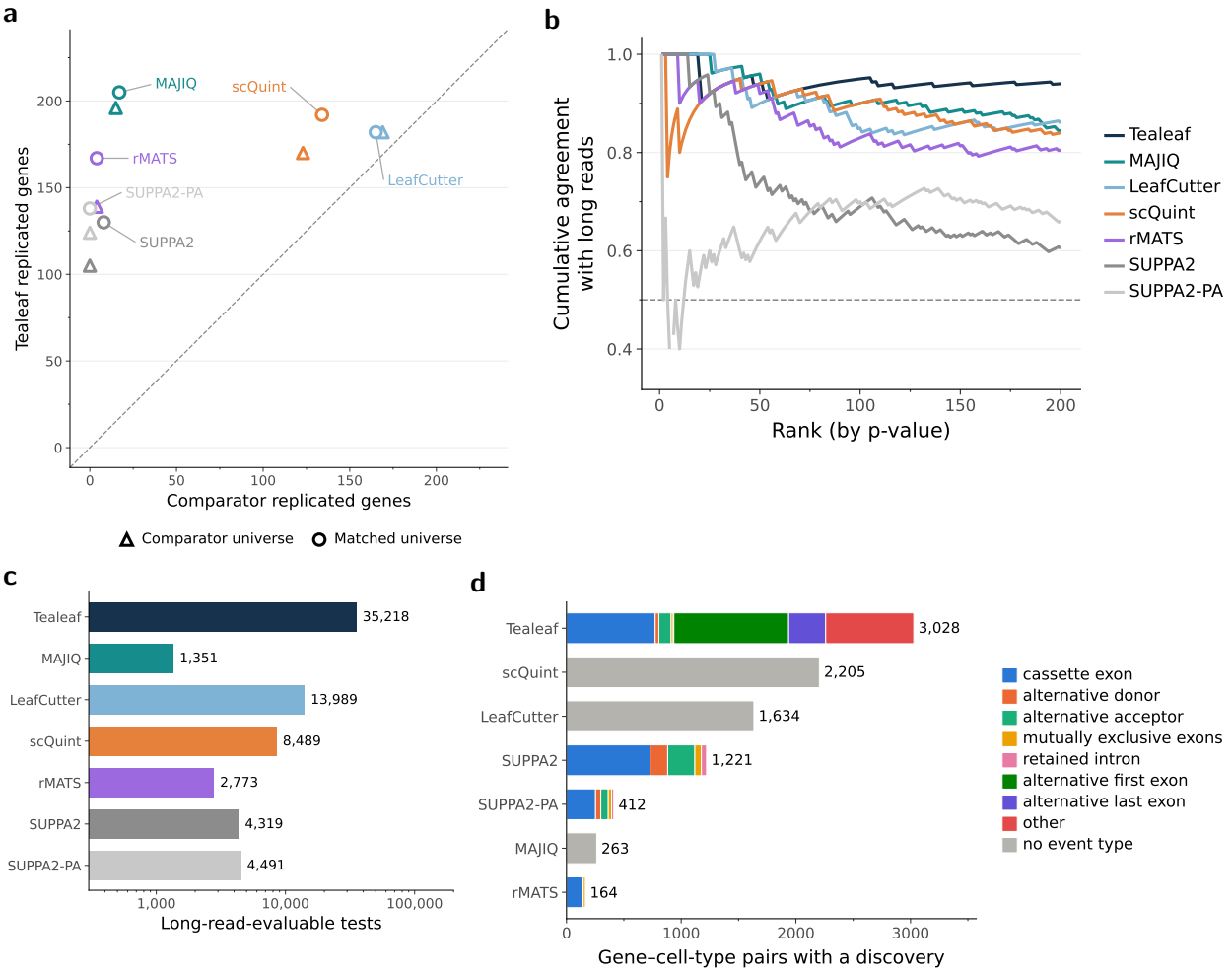


Figure 2: Mouse brain. **a**, Genes replicated across the two halves of the animals by Tealeaf and by each comparator, on the comparator's matched universe (number of genes under the name; SUPPA2-PA is primer-aware SUPPA2). **b**, Cumulative agreement with long-read directions among the top-ranked full-data tests (Tealeaf) or features (comparators); the dashed line is chance. **c**, Number of long-read-evaluable tests or features (SUPPA2 and rMATS, 500 top-ranked events per contrast). **d**, Gene–cell-type pairs with a full-data discovery (BH 0.05 within method) on the 28 pairs of the long-read cell types, by the event class of the pair's smallest-p discovery. LeafCutter, scQuint and MAJIQ test junction clusters, which have no event class. Tests called by the coverage-only null are removed for every method except SUPPA2.

exon and 40 to 51% of mutually-exclusive-exon tests at 0.05. Most rMATS mutually-exclusive-exon calls in these data have both exons in one annotated transcript, so the event compares junctions of different isoforms.

The filter keeps a test unless the null calls it, so it removes coverage-driven tests but does not recalibrate the remaining ones. It removes 0.4 to 0.9% of the recomputed tests of Tealeaf, LeafCutter and scQuint, 8 to 27% for rMATS and 43 to 96% for MAJIQ (Fig. 3b). After the filter, MAJIQ replicates 17 genes on its matched brain universe and its 100 top-ranked features agree with the long reads for 90; the results without the filter are given in Supplementary Note 4.

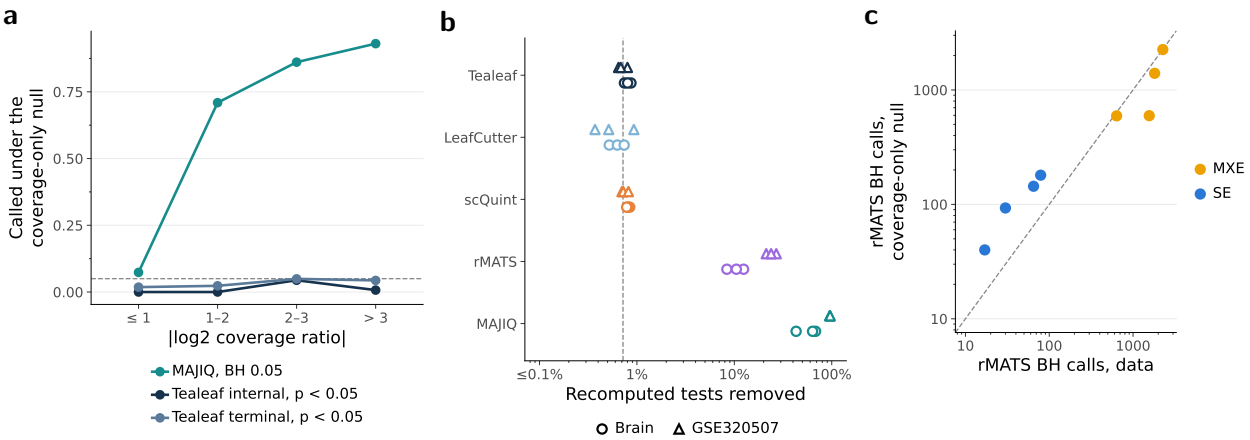


Figure 3: Coverage-only null. **a**, Fraction of tests called under the coverage-only null by the absolute $\log_2$ ratio of coverage between the two cell types in GSE320507, the human blood dataset described below: MAJIQ Heterogen edges (the junctions of its local splice variants) at BH 0.05 within contrast, and Tealeaf tests (200 per structure) at nominal $p < 0.05$ among converged fits. The dashed line is 0.05. **b**, Fraction of recomputed tests ($p < 0.01$ on the data) that the null calls (median of three null p-values below 0.05) and that are removed, per method and run (full data and the two halves), in brain and GSE320507; the dashed line is the 0.7% expected for a calibrated test. **c**, BH calls of paired rMATS on four whole GSE320507 contrasts, on the data and on counts redrawn without a difference in inclusion, for skipped exons (SE) and mutually exclusive exons (MXE).

2.5 Examples in mouse brain

We show the full-data tests with the largest effect among those evaluable in the long reads, selected without regard to long-read agreement ($p < 10^{-8}$, at least 50 long-read UMIs per cell type, at most four paths, one per gene, at least one internal and one terminal block), which all agree with the long reads in direction (Fig. 4; Supplementary Figs. S6–S8). In *Pkn2*, astrocytes include a cassette exon in 68% of molecules and cortical excitatory neurons in 27% (means over 13 animals with both cell types); the long reads show inclusion in 100% and 87% of transcripts that span the block. The direction agrees, but cortical neurons skip the exon in 73% of Parse molecules and 13% of long-read transcripts. In *Mctp1*, a terminal block with two first exons tens of kilobases apart, dentate gyrus granule neurons use the upstream first exon in 74% of junction-spanning molecules and medium spiny neurons in 15% (long-read shares 33% and 18%). The other two examples are a short cassette exon in *Nfasc* and a 54-base cassette exon in *Akap9* that is absent from the basic annotation, so the block is built from unannotated junctions (Methods, Section 4.5.2). Most Parse molecules in this block are compatible only with the precursor. Of the molecules compatible with a single mature path, all 19 in medium spiny neurons skip the exon and 14 of 15 in oligodendrocytes include it, and the fitted skipping shares are 62% and 35% (means over 15 animals with both cell types). The long reads, assigned to the comprehensive annotation, in which one transcript contains the exon, give skipping shares of 76% and 36%.

2.6 Human blood

We processed three public Parse datasets of peripheral blood mononuclear cells (PBMCs) with the same pipeline, the GRCh38 genome and the GENCODE v49 basic annotation [31]: GSE320507 [32], 38 donors with long-term COVID-19 syndrome (long COVID, LTCS; 26), after recovery from COVID-19 (CONV, 5) and healthy controls (HC, 7), and two stimulation series of 12 [33] and 8

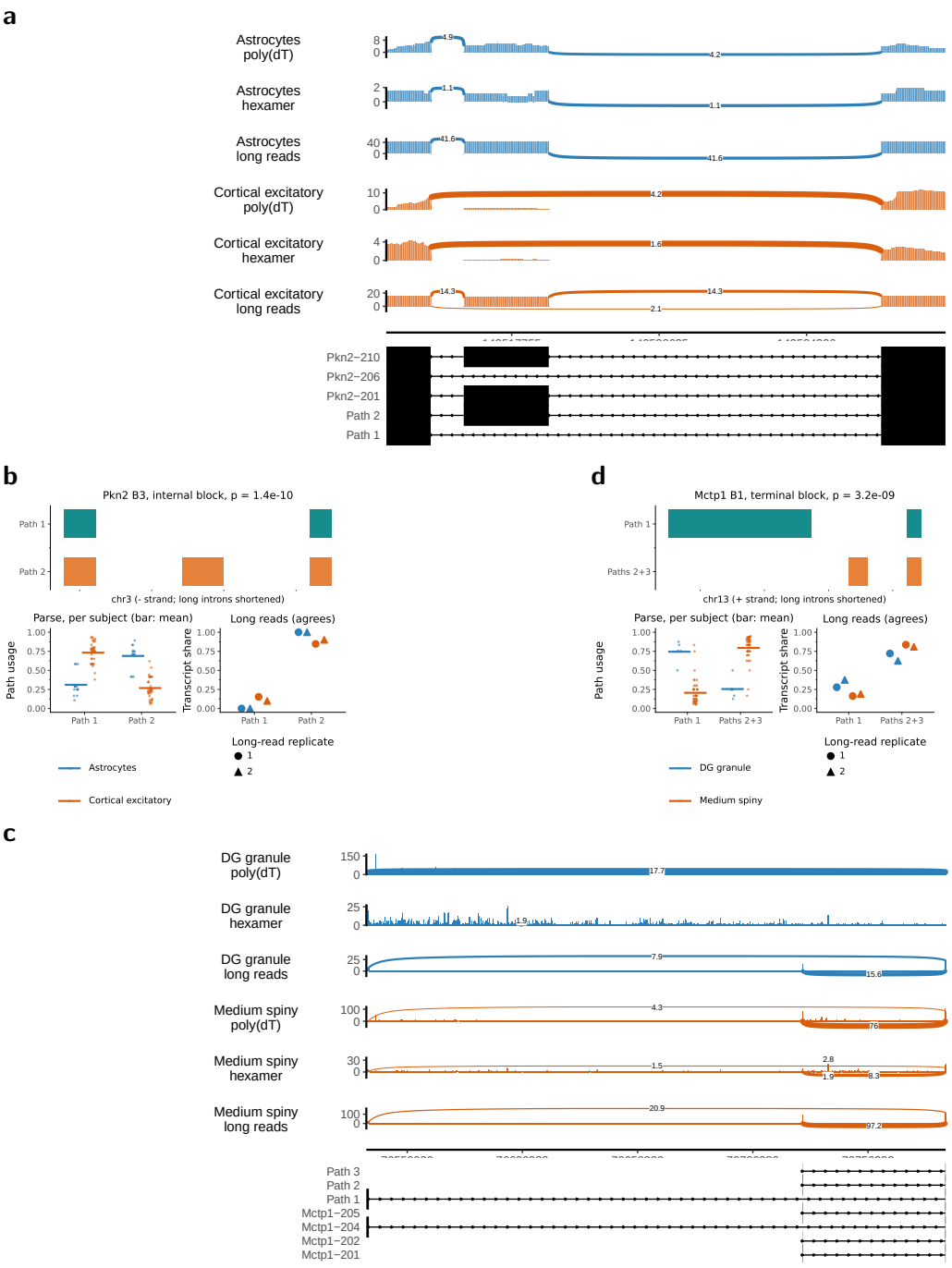


Figure 4: Examples in mouse brain. **a**, Sashimi plot of *Pkn2* block 3, a cassette exon included by path 2, in astrocytes and cortical excitatory neurons: Parse coverage from UMI-deduplicated alignments by primer, per 1,000 half-cells of the primer (each nucleus has one barcode per primer, a half-cell), and long-read coverage implied by the UMI counts of annotated transcripts, per million long-read UMIs of the cell type. Arcs are junctions labelled with their coverage, and long introns are shortened; below are the block's paths and the gene's annotated transcripts. **b**, The tested paths (top), the path usage of each animal in the Parse data with the mean over animals (bottom left) and the transcript share of each path in the two long-read replicates (bottom right). **c**,**d**, As **a**,**b** for *Mctp1* block 1, an alternative first exon in a terminal block, in dentate gyrus granule neurons and medium spiny neurons; paths 2 and 3 share their junctions and are merged in the junction-only test. Whole-gene views are in Supplementary Fig. S5.

donors [34], of which we use only the unstimulated or vehicle-control cells for cell-type contrasts. We labelled cell types with CellTypist [35] (19, 10 and 8 types). Long-read directions come from a public PacBio Kinnex library made from a Parse PBMC library of other donors, for which we called cell barcodes, UMIs and the reverse-transcription primer from the reads (93% of 148 million reads assigned to a cell; Methods).

In GSE320507, with 19 donors per half, 65 genes replicate between halves on the keys Tealeaf tests in both, and 91% of the genes discovered in one half have $p \leq 0.05$ in the other (Supplementary Table S3). Most replicated genes separate monocytes from lymphocytes. With 6 donors per half GSE327907 replicates 5 genes, and GSE334840, with 4 donors per half and 66-base reads, almost none. Of the 100 top-ranked long-read-evaluable tests, 99, 96 and 82 agree with the long reads in the three datasets (Fig. 5c; Supplementary Table S4). Tests in blood reject less often than tests in brain at the same coverage, 2.4% at $p < 0.01$ in GSE320507 against 9.0% in brain for rows with a median of 20 to 50 molecules, consistent with smaller splicing differences between blood cell types than between neuronal and glial cell types.

We ran the six comparators on GSE320507 as on the brain data, with the same coverage-only filter. In blood, LeafCutter replicates more genes than Tealeaf on its universe of 3,588 genes (126 against 67; Fig. 5a), its top-ranked intron clusters agree with the long reads as often (A_{100} 0.991 against 0.988; Fig. 5b), and it has more evaluable tests (59,097 against 44,349). scQuint replicates about as many genes as Tealeaf (67 against 54), with $A_{100} = 1.000$ over 19,875 evaluable tests. MAJIQ, after the removal of 96% of its recomputed edges, replicates 26 genes against 68. rMATS replicates 133 genes against 60, but 64 of its 100 top-ranked events agree with the long reads, so most of its replicated genes are not supported by the long reads. SUPPA2 and primer-aware SUPPA2 replicate no genes and their top-ranked events agree at about chance, although SUPPA2 makes 65,091 discoveries on the 144 pairs every method tests, against 1,903 for Tealeaf (Supplementary Fig. S4).

2.7 Tests between groups of donors

GSE320507 also allows tests between disease groups within a cell type, where the two levels are different donors. Each cell type has four contrasts, LTCS against HC, LTCS against CONV, CONV against HC, and LTCS against HC and CONV pooled, and a fifth contrast compares two sets of 26 and 12 donors drawn at random, which has no group effect. The random contrast rejects below the nominal rate at $p < 0.01$ and $p < 0.001$ and has no discoveries (Fig. 5d), so the test is not anticonservative with 12 donors in the smaller group. LTCS against HC gives 215 discoveries in 153 genes, mostly in naive and central-memory helper T cells and classical monocytes. The CONV donors share these differences. Of the 215 tests, 205 are also tested between CONV and HC, 59% with $p < 0.05$ and 92% with the same direction, while LTCS against CONV has $p < 0.05$ for 9% (3% for the random contrast). The differences therefore separate healthy donors from all donors after COVID-19, not long COVID from recovery, and may reflect other differences between the donor groups, such as age or time of sampling, which are not available. Fig. 5e shows the discovery with the largest effect under the selection rule of Methods, a 69-base cassette exon of *ANKRD12* in classical monocytes, included in 42% of HC and 75% of LTCS molecules; the other three examples are in Supplementary Figs. S9–S11.

3 Discussion

Split-pool single-cell RNA-seq with random-hexamer priming observes splice junctions in transcript bodies across many cell types and donors. Tealeaf uses these data by testing local splice paths with

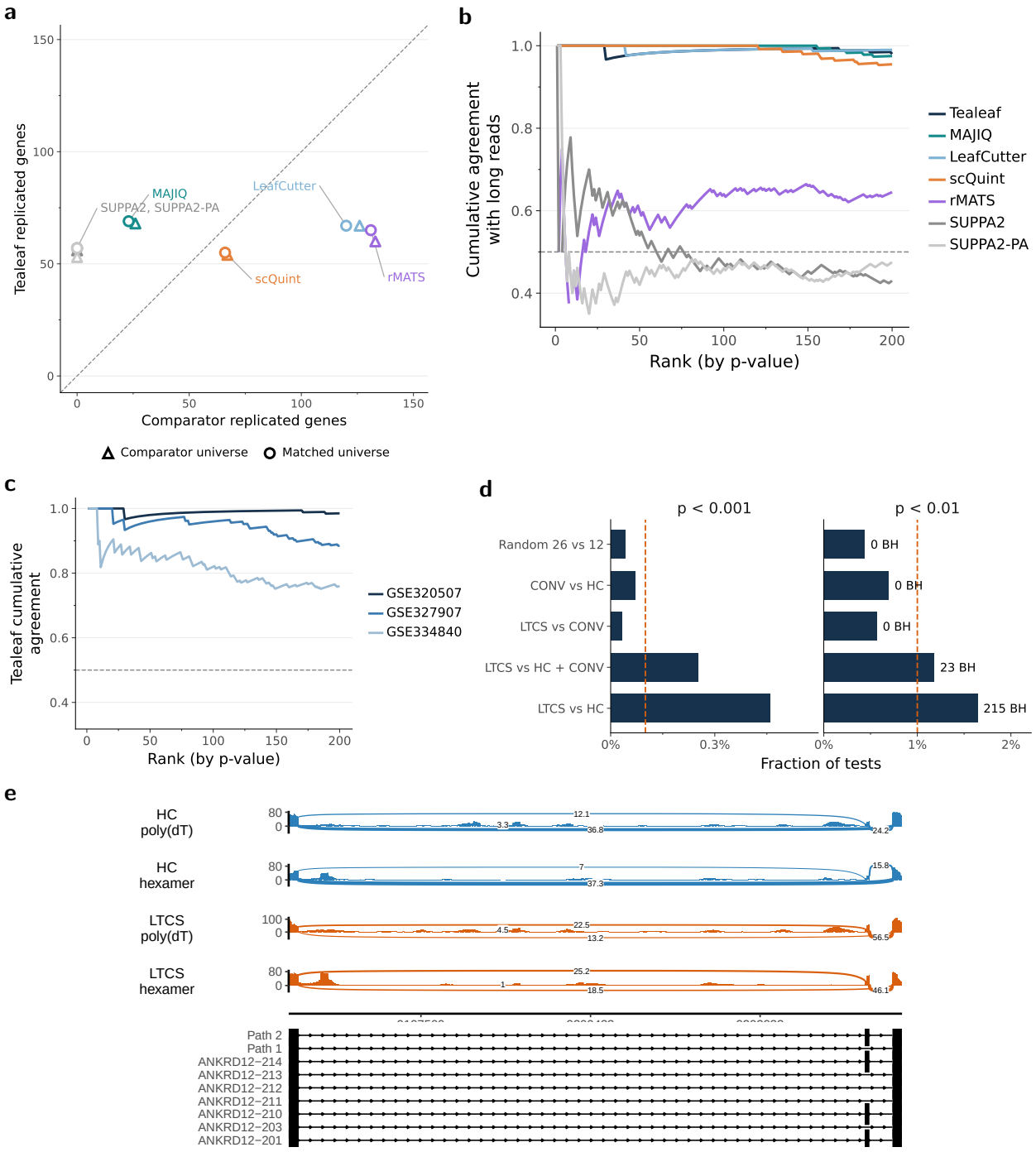


Figure 5: Human blood. **a**, Replicated genes in GSE320507 on each comparator's universe, as in Fig. 2a. **b**, Cumulative agreement with the directions of the Kinnex Parse long reads in GSE320507, as in Fig. 2b. **c**, Tealeaf cumulative agreement in the three PBMC datasets. **d**, Fraction of tests with $p < 0.001$ and $p < 0.01$ for disease-group contrasts within cell types in GSE320507, with the number of BH discoveries within each contrast; Random compares two sets of 26 and 12 donors drawn without regard to group. Dashed lines are the nominal rates. **e**, *ANKRD12* block 2 in classical monocytes of HC (7 donors) and LTCS (26 donors): Parse coverage by primer per 1,000 half-cells (cell barcodes of one primer), with the block's paths and the gene's annotated transcripts; the long-read library has no disease groups. Tests called by the coverage-only null are removed in **a** and **b** for every method except SUPPA2.

every molecule that overlaps a block, both primers in one likelihood and donors as replicates. In mouse brain it replicates more genes across halves of the animals than six comparators, on each comparator's own universe, most narrowly LeafCutter (182 against 165 genes), and its top-ranked directions agree with long reads as often as the best comparator's, over 2.5 to 26 times as many evaluable tests. Path tests cover alternative first and last exons and unannotated paths in the same framework as cassette exons, and these make up much of its discoveries. Each test names the exon chains that change and their event class, which makes a discovery easier to interpret and to annotate than a change among the junctions of a cluster.

Split replication rewards a test that responds to expression rather than splicing, because expression differences between cell types are shared by all donors and replicate across halves. Long-read direction is less affected, since such tests agree with the long reads at random, but it covers few genes. The coverage-only null addresses this for every method in the same way. It shows that most MAJIQ and many rMATS calls in these data arise from coverage alone, which a per-sample PSI estimate with a prior, or a test with too small a variance, does not separate from splicing. The filter removes such tests but cannot correct a test whose null distribution is too wide, and rMATS keeps more replicated genes than Tealeaf in blood with long-read agreement near 0.6. A comparison of splicing methods on single-cell data needs such a null, and replication alone is not sufficient.

In blood, LeafCutter replicates more genes than Tealeaf in the largest dataset, with as many long-read-evaluable tests and the same long-read agreement, and scQuint matches Tealeaf. Splicing differences between blood cell types are smaller than between neural cell types, and most of Tealeaf's discoveries in blood are at transcript ends, where it uses junction-spanning molecules only. Where a junction method performs as well, a Tealeaf test still states which exon chains change, including unannotated ones and alternative transcript ends, which a junction cluster leaves to annotation after the test. With few donors per half, as in the two stimulation series, few genes replicate for any test.

The evaluation has limits. The long-read data come from other animals or donors and cover eight brain and up to fourteen blood cell types, and the comparison is by direction only. Long reads from the same samples, with the same cell-type labels, would allow comparing effect sizes. We test intron retention separately, and with these data the test is underpowered (Supplementary Note 1). The disease contrasts in GSE320507 separate healthy donors from donors after COVID-19 and cannot be attributed to disease with the available sample information. Tealeaf's tests are calibrated in simulations on the real designs and in random contrasts between groups of donors, so they can be used for cohort designs in which donors, not cells, are the replicates.

4 Methods

4.1 Mouse brain data and notation

The brain data are split-pool barcoded single-nucleus RNA-seq (Parse Biosciences) of mouse brain, 48 subjects in six genotype or treatment groups (8 subjects each), in 8 sublibraries sequenced in 16 runs (about 345 million read pairs per run, 5.6 billion in total). We use *subject* for an animal or a human donor and call the subjects of the human data donors. Read 1 (151 bases) is cDNA in the sense orientation of the RNA. Read 2 carries the unique molecular identifier (UMI) at bases $[0, 10)$ and three 8-base barcode segments, one per split-pool barcoding round, at $[10, 18)$, $[48, 56)$ and $[78, 86)$. The first round is reverse transcription in the cell. Each cell is barcoded in one first-round well primed with poly(dT) and in one primed with random hexamers, so each cell has two *half-cell* barcodes, one per primer. We use the annotated cells of the dataset (cell-type labels from the original clustering) whose pair of half-cell barcodes belongs to that cell only, and exclude a barcode

pair shared by cells in different sublibraries, which cannot be resolved from the barcode. We use both halves of each cell, with no count threshold, giving 37,002 cells (74,004 half-cell barcodes) in 20 cell types, the cells used for the comparator methods (Section 4.11) without the shared barcode pairs. A half-cell barcode determines its subject u, cell type c and primer $p \in \{\mathrm{dT}, \mathrm{hex}\}$ (poly(dT) or random hexamer).

Throughout, intervals are 0-based and half-open, and an exon (x, y) is the interval $[x, y)$. A *junction* (d, a) is the intron $[d, a)$, where d is the end of the donor exon and a the start of the acceptor exon. A *splice block* b is a window of one gene, bounded by constitutive exons, in which transcripts take two or more exon chains, the block's *mature paths* (Section 4.5.1); every block also has a *precursor* path, the unspliced window. For a block b with S_b mature paths, path $s \in \{1, \ldots, S_b\}$ is an exon chain $E_s = ((x_{s1}, y_{s1}), \ldots, (x_{sm_s}, y_{sm_s}))$ with m_s exons, and J_s is its set of junctions $\{(y_{si}, x_{s,i+1}) : y_{si} < x_{s,i+1}\}$. The block window is $W_b = [\min_s x_{s1}, \max_s y_{sm_s})$. A compatibility class k is a non-empty subset of paths, encoded as a bit mask; a molecule's class is the set of paths compatible with its reads (Section 4.6).

4.2 Targeted reference and index

Target loci. The targets are the 33,755 genes with a multi-exon annotated transcript, so that genes without an annotated block of several paths can still receive blocks from unannotated junctions (Section 4.5.2). Each gene span (GENCODE vM32 basic annotation, GRCm39), from the first to the last base of its annotated transcripts, is a target interval, and we merge overlapping intervals into a locus. We do not pad the spans, since novel junctions are used only inside annotated gene spans (Section 4.5.2), so flanking sequence contributes no path, and padding each span by 1 kb changes fewer than 0.2% of the molecules counted in blocks (Section 4.6). This gives 26,612 loci with 1.19 Gb of sequence. The reference is the concatenation of the locus sequences and the spliced sequences of the 59,378 annotated transcripts of the target genes, separated by runs of 300 N, in a 2-bit code with N as a fifth symbol.

Index. We index every k-mer ($k = 23$) without N that starts in a locus, and every transcript k-mer that spans an exon junction of its transcript. We sort the k-mer codes, and a table over the leading 24 bits of the code gives the range of each prefix. The index holds 1.19 billion k-mers.

Genome uniqueness. The aligner sees only the target loci, so a read from a paralog or repeat elsewhere in the genome can align to a locus with a good score. We replace the genome-wide search for competing alignments by a uniqueness flag per indexed k-mer. We look up the canonical code (the smaller of the code and the code of its reverse complement) of every k-mer of the genome in the index and count occurrences. A locus k-mer is *genome-unique* if its canonical code occurs exactly once in the genome; a transcript junction k-mer is genome-unique if it does not occur in the genome. 83.7% of indexed k-mers are genome-unique. We accept an alignment as unique only if one of its aligned blocks (the gap-free segments of the alignment) contains a k-mer that matches the read exactly and is genome-unique. A read from another genomic copy cannot contain such a k-mer unless a sequencing error creates it.

4.3 Read alignment

We align each read $q \in \{0, 1, 2, 3, 4\}^L$ ($L = 151$; the four bases coded 0 to 3 and N as 4) in both orientations, q and its reverse complement $\bar{q}$, and all candidate alignments of both orientations compete. Algorithm 1 gives the procedure for one orientation, and the paragraphs after it define

its extension, overhang, scoring and status rules. Scores follow STAR [23]. For an alignment with $M \in \mathbb{N}_0$ matching and $X \in \mathbb{N}_0$ mismatching bases (read N bases are neutral), aligned blocks separated by junctions $j \in \mathcal{J}$ and indels $i \in \mathcal{D}$ of length g_i, and reference span ℓ (from the first to the last aligned base, introns included),

$$\text{score} = M - X + \sum_{j \in \mathcal{J}} \sigma(j) - \sum_{i \in \mathcal{D}} (2 + 2|g_i|) + \min\{0, \lceil -0.25 \log_2 \ell - 0.5 \rceil\},$$

where the junction score is $\sigma(j) = +2$ for a known junction (in the first pass an annotated junction, see *Two passes* below), 0 for an unannotated GT/AG (or CT/AC) junction, -4 for GC/AG (CT/GC) and -8 for AT/AC (GT/AT); these are the canonical motifs. A gap of at least 21 bases between consecutive aligned blocks is a junction; shorter gaps of at most 6 bases are deletions (positive gap) or insertions (negative gap).

Algorithm 1 Alignment candidates of one read orientation q of length L.

1: **Seeds.** For read positions $r = 0, 4, 8, \ldots$ and $r = L - k$, look up the k-mer $q[r, r + k)$ unless it contains N; keep hits of k-mers with at most 32 index occurrences (at most 1,024 hits). Each hit at reference position t has diagonal $\delta = t - r$.
2: **Segments.** Merge hits with equal δ into segments $(\delta, r_{\text{start}}, r_{\text{end}})$ spanning the read positions of their k-mers.
3: **for** each segment on a transcript **do**
4: Extend ungapped along the transcript in both directions (X-drop, at most 10 mismatches in total); project the aligned read interval onto the transcript's exons to give genomic blocks; add as a candidate.
5: **end for**
6: **Chaining.** Sort locus segments by r_{start}. Dynamic programming, $D_j = r_j^{\text{end}} - r_j^{\text{start}}$, and for each predecessor i in the same locus with $r_i^{\text{start}} < r_j^{\text{start}}$, $r_i^{\text{end}} \le r_j^{\text{end}}$ and diagonal gap $g = \delta_j - \delta_i \in [-6, -1] \cup [1, 6] \cup [21, 10^6]$, $D_j \leftarrow \max\{D_j, D_i + r_j^{\text{end}} - \max(r_j^{\text{start}}, r_i^{\text{end}})\}$.
7: **for** each chain end in decreasing D, not yet used, with $D \ge k$ **do**
8: Trace the chain back; extend its first segment to the left (X-drop, at most 10 mismatches).
9: **for** consecutive segments A, B of the chain **do**
10: Choose the split s within 4 bases of the segment boundary that minimizes mismatches of $q[\cdot, s)$ on δ_A and $q[s, \cdot)$ on δ_B; for a junction ($g \ge 21$) require a known junction or a canonical motif, and break ties by known, then GT/AG, then other canonical motifs.
11: **end for**
12: Extend the last segment to the right with the remaining mismatch budget.
13: **Rescue.** If the right end is soft-clipped by at least 3 bases, try junctions from split points s within 6 bases of the clip to every known acceptor a at intron length 21 to 10^6; an unknown junction needs a canonical motif and a tail of at least 5 bases. Accept the best split whose tail has at most $\max(1, \lfloor (L - s)/10 \rfloor)$ mismatches and whose score gain over clipping is positive. Do the same for a clipped left end and every known donor.
14: Add the alignment as a candidate.
15: **end for**

X-drop extension. An extension from a seed boundary scores $+1$ per match and -1 per mismatch, stops when the score falls 8 below its maximum, at an N in the reference, at the locus or transcript boundary, or at the mismatch budget, and ends at the position of the maximum. We soft-clip the unaligned remainder.

Overhang minimum and candidate scoring. A read's *overhang* on a junction is the length of the shorter of the two aligned blocks that flank it. Before scoring, we remove a first or last aligned block that is shorter than the minimum overhang of its junction, as with STAR's `alignSJDBoverhangMin` and `alignSJoverhangMin`, 3 bases for a known junction and 5 for an unannotated one. We merge candidates with identical reference start, end, aligned-block count and orientation, keeping the higher score. A table holds at most 24 candidates; when it is full, a new candidate replaces the lowest-scoring one if it scores higher.

Read status. Let $c^\star$ be the highest-scoring candidate over both orientations. The read is

1. *too short* if $\text{score}(c^\star) < \lfloor 0.66(L-1) \rfloor$, $M(c^\star) < \lfloor 0.66(L-1) \rfloor$, $X(c^\star) > 10$ or $X(c^\star) > 0.3(M+X)$;
2. *multi-mapping* if another candidate scores at least $\text{score}(c^\star) - 1$ and does not overlap $c^\star$ on the reference, or lies on the other orientation;
3. *not genome-unique* if no aligned block of $c^\star$ contains an exactly matching genome-unique k-mer;
4. *unique* otherwise.

We use only unique alignments and map the aligned blocks of $c^\star$ back to genomic coordinates.

Two passes. The first pass uses the annotated junctions of the target transcripts as known junctions. For each sequencing run, we count the junctions of unique alignments and record the largest overhang among their reads. We keep every annotated junction, and an unannotated one if its largest overhang is at least 12 bases and it has at least 4 reads, or 3 reads and an intron of at most 200 kb, 2 reads and at most 100 kb, or 1 read and at most 50 kb (STAR's `outSJfilter` defaults for canonical motifs). The intron strand is the strand of the motif. We sum counts over runs and record, for each junction, the number of sublibraries with reads. These counts use every uniquely aligned read, irrespective of its barcode, and never use cell-type labels. The second pass adds to the known junctions every first-pass junction with a canonical motif, at least 20 reads and at least 4 of the 8 sublibraries, the support required of a novel path (Section 4.5.2); junction counts from the second pass then define the novel paths. Known junctions receive the +2 score, the 3-base overhang minimum and the rescue of clipped ends, as in STAR's two-pass mode.

4.4 Cell barcodes, UMIs and molecules

We correct each 8-base barcode segment of read 2 to its 96-entry whitelist of valid barcodes, by replacing it with the unique nearest whitelist entry within Hamming distance 2 (N read as T); a segment without such an entry is uncorrectable. We assign a read to a half-cell barcode if all three segments correct and the triple is the barcode of a retained cell in its sublibrary. Its UMI is valid if it contains no N and is not a homopolymer. We keep unique alignments with an assigned barcode and a valid UMI.

Within each (half-cell barcode, locus), we join UMIs at Hamming distance 1 transitively and replace them by the smallest UMI of their component (the `1MM_All` rule of STARsolo [24]). A *molecule* is a (half-cell barcode, collapsed UMI, block) triple, over the reads of both sequencing runs of a sublibrary.

4.5 Splice blocks and paths

4.5.1 Annotated blocks

Annotated splice blocks are local regions between constitutive exonic anchors in the annotation. We collapse transcripts of a gene that take the same exon chain through the region into one path, and a block has the paths of all annotated transcripts that traverse it. Each path is the chain left anchor exon, internal exons, right anchor exon, with exons that touch merged into one, so transcripts whose chains differ only by an exon boundary inside a continuous exonic interval share a path. Every block also carries the precursor path, the single exon W_b, which is compatible with reads without a junction inside the window (intronic or exon-body reads). A path whose chain is the whole window duplicates the precursor path, and we remove it. We use blocks of the target genes with 2 to 30 annotated paths.

A block is *internal* if all its annotated paths start with the same exon and end with the same exon, and *terminal* otherwise (alternative first or last exons, transcript starts or ends). We decide the structure from the annotated paths only, so adding novel paths does not change it.

4.5.2 Novel paths

We build novel paths from the second-pass junction support. For a block b, let $\mathcal{A}_b = \bigcup_s J_s$ be its annotated junctions, $R(j) \in \mathbb{N}_0$ and $\Lambda(j) \in \mathbb{N}_0$ the read count and number of sublibraries of junction j, and $R_b^\star = \max_{j \in \mathcal{A}_b} R(j)$. A junction $(d, a) \notin \mathcal{A}_b$ is a *novel junction* of b if $d, a \in W_b$, its intron strand is the gene's strand, and

$$R(d,a) \geq \max\{20,\, 0.02 R_b^\star\}, \qquad \Lambda(d,a) \geq 4.$$

For an annotated path E with exons $(x_1, y_1), \ldots, (x_m, y_m)$, the donor d attaches to exon i of E if $x_i < d \leq y_i$ (an earlier donor within the exon) or $y_i < d < x_{i+1}$ with $d - y_i \leq 300$ (an exon extension into the intron); the acceptor a attaches to exon j if $x_j \leq a < y_j$ or $y_{j-1} < a < x_j$ with $x_j - a \leq 300$. Novel junctions modify one annotated path each, by the rules of Algorithm 2 and Supplementary Fig. S2. We keep at most 8 novel paths per block, ordered by the smallest read count of their novel junctions, and discard a novel path identical to an annotated one.

New blocks. Novel junctions inside a gene that lie in no annotated block window, for example a junction that skips an exon constitutive in the annotation, define new blocks. We use such a junction if it satisfies the support criteria above, with $R^\star$ the largest read count of the annotated junctions overlapping it, and cluster used junctions with overlapping introns. For a cluster with first donor d and last acceptor a, each annotated transcript gives the segment of its exon chain from the exon to which d attaches to the exon to which a attaches; if only one end attaches, the segment starts at the last exon ending before d or ends at the first exon starting after a, which allows novel first and last exons. We clip segments to their shared extent; distinct clipped segments are the annotated paths of the new block, and Algorithm 2 adds its novel paths. We keep a new block if it has at least one novel path.

Block set. On the second pass the block set has 34,677 blocks, 22,898 annotated and 11,779 new, with 21,509 novel paths.

Algorithm 2 Novel paths of block b from its novel junctions $\mathcal{N}_b$; $T = 151$ is the new terminal exon length.

Pairs. For novel junctions $(d, x), (y, a) \in \mathcal{N}_b$ with $d < x < y < a$ and $y - x \leq 1{,}000$, form the candidate (d, a) with new exon $[x, y)$; mark both junctions as paired.
Singles. Every $(d, a) \in \mathcal{N}_b$ is also a candidate without a new exon.
for each candidate and each annotated path E **do**
    if d attaches to exon i and a to exon $j \geq i$ of E **then**
        New chain: exons $1..i-1$ of E, (x_i, d), the new exon if any, (a, y_j), exons $j+1..m$ of E. A single that is paired may not use an exon extension.
    else if the candidate is an unpaired single, d attaches to no exon of E and a attaches to exon j **then**
        New first exon: $(\max\{d - T,\, y^-\}, d)$, then (a, y_j) and exons $j+1..m$, where y^- is the largest exon end of E before d (or the window start).
    else if the candidate is an unpaired single, d attaches to exon i and a to no exon of E **then**
        New last exon: exons $1..i-1$, (x_i, d), then $(a, \min\{a+T,\, x^+\})$, where x^+ is the smallest exon start of E after a (or the window end).
    end if
end for
For each candidate, keep the chain from the path where it replaces a junction sharing its donor or acceptor, then with the least exon extension, then whose least supported junction has most reads.

4.6 Molecule classes

Read compatibility. Consider a read with aligned blocks $\{[\alpha_r, \beta_r)\}$, indexed by r, and junctions $\mathcal{J}_{\text{read}}$, and a block b whose window it overlaps. We clip the read's aligned blocks to W_b and restrict its junctions to those inside W_b. The read is compatible with path s (or the precursor path) if every clipped aligned block lies within one exon of E_s and every junction inside the window belongs to J_s. The read's class in block b is the mask of compatible paths. We leave a read of class 0, compatible with no path (for example a read with an unannotated junction not represented by a novel path, or with an aligned block reaching into an intron), out of the block. A molecule's class is the bitwise AND of the classes of its remaining reads that overlap the window; we discard a molecule whose reads are compatible with disjoint sets of paths, which has class 0. A read enters every block of its locus whose window it overlaps.

Junction-only classes. A junction is *variable* in block b if it belongs to some but not all mature paths. We count molecules in two collections. The *all-read* collection keeps every molecule of non-zero class, and the *junction-only* collection keeps a molecule only if at least one of its reads of non-zero class has a variable junction inside the window.

Counts. For each block, we sum molecule counts over the half-cell barcodes of each (subject, cell type, primer) and over sublibraries, giving $y_{ucpk} \in \mathbb{N}_0$ for subject u, cell type c, primer p and class k. Internal blocks use the all-read counts and terminal blocks the junction-only counts.

4.7 Read opportunities

Read geometry determines the probability of each class given the path composition. For path s with spliced length $\lambda_s = \sum_i (y_{si} - x_{si})$, we consider a read of length $L = 151$ starting at every position $t \in \{1 - L, \ldots, \lambda_s - 1\}$ of the path's spliced sequence, so that it overlaps the path by at least one base, and clip the part outside the path, as reads leaving the window are clipped in the data. We map the read's spliced interval to genomic aligned blocks and junctions and classify it with the same compatibility rule. The opportunity $n_{ks} \in \mathbb{N}_0$ is the number of start positions on path s that give class k, and $L_s = \sum_k n_{ks}$ is the effective length of path s. For junction-only classes, only start positions whose read spans a variable junction count. We compute opportunities for the S_b mature paths and the precursor path.

Under a model in which molecules of path s arise in proportion to an abundance $\rho_s \geq 0$ per position and start uniformly along it, $P(\text{class } k) = \sum_s \rho_s n_{ks} / \sum_s \rho_s L_s$. The molecule share of path s is $\pi_s = \rho_s L_s / \sum_{s'} \rho_{s'} L_{s'}$.

4.8 Block test

4.8.1 Projection and screens

A test is indexed by a block b and an unordered pair of cell types $\{c_0, c_1\}$, with c_0 the first in alphabetical order; we report effects (defined at the end of this section) as c_1 minus c_0. The data of a test are the counts y_{ucpk} for $c \in \{c_0, c_1\}$.

1. **Eligibility.** At least 4 subjects have molecules in both cell types, and the block has at least 20 molecules in the two cell types.

2. **Same-junction groups (terminal blocks).** Paths with identical junction sets cannot be separated by junction-spanning reads, and we merge them into one group. A class is compatible with a group if it is compatible with any member, and the group opportunity is the mean of its members' opportunities. A block with fewer than two groups is not tested.

3. **Share screen.** We estimate molecule shares $\widehat{\boldsymbol{\pi}}$ from the class counts pooled over both cell types, both primers and all subjects, by the expectation–maximization (EM) algorithm for $P(k) = \sum_s \pi_s n_{ks} / L_s$, with a pseudocount of 0.5 molecules per path per iteration (tolerance 10^{-10}, at most 1,000 iterations). For internal blocks the precursor path is a component of the EM, and we remove it afterwards and renormalize the mature shares. We remove paths with $\widehat{\pi}_s < 0.01$. A block with fewer than two remaining paths is not a hypothesis and leaves the family of tests to which we apply multiple-testing correction (*Hypothesis family* below). The screen uses no cell-type labels within the pair.

4. **Projection.** We restrict masks to the retained mature paths and drop the precursor bit. We discard molecules compatible only with the precursor or only with removed paths, and merge classes that become identical, summing their opportunities. We then apply the eligibility criteria again to the projected counts.

After projection let S be the number of retained paths and K the number of classes.

4.8.2 Likelihood of one path against the rest

For target path s, we collapse the other paths into one component with weights

$$w_j = \frac{\widehat{\pi}_j / L_j}{\sum_{j' \neq s} \widehat{\pi}_{j'} / L_{j'}}, \qquad j \neq s,$$

the per-position abundances implied by the pooled molecule shares, so that the collapsed component's molecules divide among its paths in proportion to $\widehat{\pi}_j$. For class k let

$$\beta_k = n_{ks}, \qquad \eta_k = \sum_{j \neq s} w_j n_{kj}, \qquad B = \sum_k \beta_k, \qquad H = \sum_k \eta_k.$$

We keep all classes, including the class compatible with all paths, whose relative frequency informs ψ through coverage. For a subject–cell-type row (u, c) with target usage $\psi \in (0, 1)$, the class probabilities in primer p are

$$f_{pk}(\psi) = \frac{\psi_p \beta_k + (1 - \psi_p)\eta_k}{\psi_p B + (1 - \psi_p)H}, \qquad \psi_p = \operatorname{expit}(\operatorname{logit} \psi + \gamma_p),$$

with $\gamma_{\mathrm{dT}} = 0$ and $\gamma_{\mathrm{hex}} = \gamma$, a primer offset that absorbs differences in positional capture between the primers. ψ is the target's share of per-position abundance (the ρ scale of the read-opportunity model), so its expected molecule share in primer p is $\psi_p B / \{\psi_p B + (1 - \psi_p)H\}$. Conditioning on the primer totals, the row log-likelihood is

$$\ell_{uc}(\psi) = \sum_{p \in \{\mathrm{dT}, \mathrm{hex}\}} \sum_{k=1}^{K} y_{ucpk} \log f_{pk}(\psi).$$

Primer offset. We estimate γ before the test without cell-type labels, by maximizing the profile likelihood in which every row has its own usage,

$$\widehat{\gamma} = \arg \max_{\gamma \in [-4,4]} \left\{ \sum_{(u,c)} \max_{t \in \mathbb{R}} \left[\ell_{uc,\mathrm{dT}}(\operatorname{expit} t) + \ell_{uc,\mathrm{hex}}(\operatorname{expit}(t + \gamma)) \right] - \frac{\gamma^2}{2\tau^2} \right\},$$

over rows observed with both primers, where $\ell_{uc,p}$ is the primer-p term of ℓ_{uc}. The maximization uses a grid of step 0.05 in γ and in $t \in [-22, 22]$, refined by a parabola through the three best points. We estimate $\widehat{\gamma}$ for each target path and hold it fixed in both hypotheses. The penalty is a Gaussian prior on γ with mean 0 and standard deviation $\tau = 1$ on the logit scale. Without it, the profile likelihood is often flat, because the rows of two cell types hold few molecules of one primer in the classes that separate the paths, and in the brain data 71% of the estimates lie at the bounds ± 4, against 0.2% with the prior. With the same blocks, tests and scoring, the prior increases the converged tests from 646,782 to 664,275 and the genes replicated between halves on the matched universes by 1 to 13 for the six comparators, leaves the long-read A_{100} unchanged (0.946 and 0.947), and brings the fitted path usage closer to the long-read path shares (mean absolute difference 0.237 and 0.203, Pearson correlation 0.57 and 0.66, over 19,670 two-path tests that the long reads can evaluate).

Internal poly(dT) priming. Poly(dT) primers also anneal to A-rich sequence inside transcripts. On 3,000 long single-isoform genes per library, we count the 3′ ends of reads relative to the A-runs (at least 8 A in 10 bases of the sense sequence) of the spliced transcript, against a uniform placement of the same reads. Poly(dT) read ends are 4.6 times as frequent as expected 0 to 100 bases upstream of an A-run, 2.3 times at 100 to 200 bases, 1.9 times 0 to 200 bases downstream of the A-run start (reads that cover the A-run) and close to 1 beyond 300 bases; random-hexamer read ends show no enrichment (0.86 to 1.06). This capture depends on each path's sequence and not on the cell type, so it changes the poly(dT) class probabilities of a block in the same way in both cell types of a

test. Blocks whose alternative exon contains an A-run have a median offset estimate of -4 without the prior, the poly(dT) molecules overrepresenting the inclusion path. We also tested poly(dT) opportunities in which each start position is weighted by one plus the excess of this enrichment profile summed over the A-runs of the path, with the same offset prior. The weighting changed the replicated genes by -5 to 0 and A_{100} by 0.001 relative to the offset prior alone, and we do not use it.

4.8.3 Random-effects model

Let $z_c = \mathbf{1}[c = c_1] \in \{0, 1\}$. Usage varies between subjects through a subject intercept and between rows through a Beta distribution,

$$v_u \sim \mathcal{N}(0, \sigma^2), \qquad \mu_{uc} = \operatorname{expit}(\theta_0 + \theta_1 z_c + v_u), \qquad \psi_{uc} \mid v_u \sim \operatorname{Beta}\big(\kappa\mu_{uc}, \kappa(1-\mu_{uc})\big),$$

with fixed effects $\theta_0, \theta_1 \in \mathbb{R}$, subject standard deviation $\sigma > 0$ and row concentration $\kappa > 0$. The subject intercept is shared by the subject's two cell types, which makes the test paired. Subjects observed in only one cell type contribute to θ_0, σ and κ. The marginal likelihood is

$$\mathcal{L}(\theta_0, \theta_1, \kappa, \sigma) = \prod_u \int_{\mathbb{R}} \phi(v; 0, \sigma^2) \prod_{c \in \mathcal{C}_u} \int_0^1 e^{\ell_{uc}(\psi)} \operatorname{Beta}\big(\psi; \kappa\mu_{uc}(v), \kappa(1-\mu_{uc}(v))\big)\, d\psi\, dv,$$

where $\phi(\cdot; 0, \sigma^2)$ is the normal density, $\operatorname{Beta}(\psi; \cdot, \cdot)$ the Beta density, and $\mathcal{C}_u$ the set of cell types observed for subject u.

Levels in different subjects. The two levels of a test can also be groups of different subjects within one cell type, such as disease groups (Section 4.13). Each subject then has one row, and $z_u \in \{0, 1\}$ indicates the subject's group. The model and test are unchanged; the subject intercept and the row Beta distribution both describe variation between subjects of a group. Eligibility requires a minimum number of subjects with molecules in each level instead of in both.

Numerical integration. We evaluate the inner integral in $t = \operatorname{logit}\psi$ on $[-18, 18]$ by the trapezoid rule and add the tails beyond ± 18 analytically at the boundary value of ℓ_{uc}. The grid spacing for row (u, c) is $\min\{0.03,\, s_{uc}/1.5\}$, where s_{uc} is a quarter of the width of the region where $\ell_{uc}(\operatorname{expit} t)$ is within 2 of its maximum. The outer integral uses adaptive Gauss–Hermite quadrature with 9 nodes, centred on each subject's posterior mode $\widehat{v}_u$ (damped Newton iterations) and scaled by the inverse square root of the negative second derivative of the log integrand at the mode.

Estimation. We find maximum-likelihood estimates by the L-BFGS-B algorithm with analytic gradients in the coordinates $(\theta_0, \theta_1, \log\kappa, \log\sigma)$, within the bounds $-12 \le \theta_0, \theta_1 \le 12$, $0.1 \le \kappa \le 2000$ and $0.01 \le \sigma \le 3$. The null fit ($\theta_1 = 0$) starts from least-squares coefficients on the rows' maximum-likelihood usages, $\kappa = 20$ and $\sigma = 0.5$; the alternative starts from the null solution, and we refit it from $\theta_1 = 0$ if it ends below the null log-likelihood.

Test. With $(\widehat{\theta}_0, \widehat{\theta}_1, \widehat{\kappa}, \widehat{\sigma})$ the alternative and $(\widetilde{\theta}_0, 0, \widetilde{\kappa}, \widetilde{\sigma})$ the null estimates, the likelihood-ratio test (LRT) statistic is

$$T_s = 2\left\{\log\mathcal{L}(\widehat{\theta}_0, \widehat{\theta}_1, \widehat{\kappa}, \widehat{\sigma}) - \log\mathcal{L}(\widetilde{\theta}_0, 0, \widetilde{\kappa}, \widetilde{\sigma})\right\},$$

and $p_s = P(\chi^2_1 > T_s)$. We re-evaluate both optima at half the ψ grid spacing and 19 quadrature nodes; if the log-likelihoods or the statistic change by more than 10^{-3}, we refit the test at the finer setting and check it again at quarter spacing and 39 nodes. We call a test converged if both fits succeed and the change is at most 10^{-3}; the reported statistic is $\max(T_s, 0)$ at the finer setting. A path test that does not converge has $p_s = 1$.

4.8.4 Block p-value and effect

For $S = 2$ we test path 1 only. For $S > 2$, we test each of the S paths against the rest, and the block p-value is the Simes combination [22]

$$p_b = \min\left\{1,\ \min_{i=1,\dots,S} \frac{S\, p_{(i)}}{i}\right\},$$

where $p_{(1)} \le \cdots \le p_{(S)}$ are the ordered path p-values. The effect for $S = 2$ is $\bar{m}_{c_1} - \bar{m}_{c_0}$ for path 1 (and its negative for path 2), where $\bar{m}_c = \int \mathrm{expit}(\widehat{\theta}_0 + \widehat{\theta}_1 z_c + v)\phi(v; 0, \widehat{\sigma}^2)\, dv$ is the marginal usage on the poly(dT) scale. For $S > 2$, we estimate the molecule-share composition of each row by the EM algorithm of the share screen with primers pooled, average the compositions over the subjects observed in both cell types (over the subjects of each level when the levels are groups of different subjects), and take the effect as the difference of the averages.

4.8.5 Hypothesis family

Internal-block tests and terminal-block tests form one family, and we apply the Benjamini–Hochberg (BH) procedure [28] over all tests of a cohort, the full data or one half of the subjects (Section 4.10). A test that fails to converge, or has fewer than four subjects in its fit, enters with $p = 1$; a block removed by a screen does not enter.

4.9 Calibration

We check calibration by count simulations under the null on real designs. For randomly chosen blocks and cell-type pairs of the full data (in brain, 200 internal and 800 terminal tests; in blood, 200 of each), we keep the observed molecule totals per (subject, cell type, primer) and redraw class counts from compositions in which the two cell types share the same path usage. We draw a subject composition from a Dirichlet distribution with concentration 20 around the pooled shares, the row composition from a Dirichlet with concentration $\varsigma \in \{5, 20, \infty\}$ around the subject composition, and class counts from the multinomial implied by the opportunities, two draws per test. For internal blocks a fourth scenario multiplies the precursor share of the rows of one cell type by 4, which the precursor component should absorb. Internal tests reject 0 to 2.1% of converged fits at nominal 0.05 and at most 0.7% at 0.01 in brain, and at most 1.7% and 0.4% in blood. Junction-only terminal tests reject 3.3–4.0% (two-path) and 2.0–2.8% (multi-path) at 0.05 in brain, at most 0.8% at 0.01 and at most 0.4% at 0.001; in blood they reject 1.6–2.6% (multi-path) and 2.2–6.5% (two-path, about 85 converged fits per scenario) at 0.05 and at most 1.1% at 0.01.

4.10 Evaluation in mouse brain

4.10.1 Subject-split replication

We divide the 48 subjects into two halves of 24 with 4 subjects of each group in each half (subjects shuffled within group and assigned alternately), and fit Tealeaf and every comparator (LeafCutter,

MAJIQ Heterogen, scQuint, rMATS, SUPPA2 and primer-aware SUPPA2; Section 4.11) separately on each half. For each comparator m, the universe $\mathcal{K}_m$ is the set of (gene, cell-type pair) keys with a finite p-value for m in both halves, taken from the comparator's own results, over up to 190 cell-type pairs; no Tealeaf result enters it. SUPPA2 and primer-aware SUPPA2 use exact paired signed-rank tests over the full event catalogue. We fit Tealeaf on every key of these universes. We score each comparison on two universes, the *matched* universe $\mathcal{U}_m = \mathcal{K}_m \cap \mathcal{T}_1 \cap \mathcal{T}_2$, with $\mathcal{T}_h$ the keys with a Tealeaf test in half $h \in \{1, 2\}$, and the *comparator* universe $\mathcal{K}_m$. In each half h, we combine the block p-values of a (gene, pair) key by Simes, give keys of the universe without a test $p = 1$, and combine the key p-values of a gene by Simes into p_{gh}. We aggregate comparator feature p-values in the same way, for LeafCutter intron clusters, MAJIQ Heterogen edges (the connections of a local splice variant, LSV), scQuint intron groups and SUPPA2 and rMATS events. Before both evaluations we remove, for every method except SUPPA2, the tests that the coverage-only null calls; this null recomputes each test on counts with the observed coverage and no difference between the two cell types (Section 4.12). The conjunction p-value $p_g^{\text{rep}} = \max(p_{g1}, p_{g2})$ tests for signal in both halves, and a gene is *replicated* if its BH-adjusted conjunction p-value over the genes of the universe is at most 0.05; we score both methods on the same universe. Replication does not require agreement of direction.

4.10.2 Long-read direction

We compare the directions of full-data tests with the P56 mouse brain single-cell long-read data of Joglekar et al. [29] (ScISOr-Seq2), using its UMI-filtered annotated-transcript count matrix. We match transcript names to the comprehensive GENCODE vM32 annotation, which gives each counted transcript an exon chain. Eight cell types have a prespecified correspondence, astrocytes (with Bergmann glia), mature oligodendrocytes, cerebellar granule neurons, cortical excitatory neurons, dentate gyrus and CA excitatory neurons, thalamic excitatory neurons and medium spiny neurons; we pool glial samples over regions within replicate. We test the full data on all pairs of these eight cell types in addition to the keys of the split-replication universes. A transcript counts for a tested path if its exon chain is compatible with that path and no other, by the rule used for reads (Section 4.6), that is, its exons clipped to the block window lie within exons of the path and its junctions inside the window are junctions of the path. We do not count transcripts compatible with several paths, which do not cover the region where the paths differ, or transcripts compatible with none. A novel path is evaluable when an annotated transcript of the comprehensive annotation has its structure. A test is evaluable if every tested path has at least one counted transcript, both cell types have at least 20 counted long-read UMIs, and both effects are finite and non-zero; 35,233 tests are evaluable before the coverage-only filter. With long-read counts $\boldsymbol{n}_c \in \mathbb{N}_0^S$ of the paths in cell type c (summed over replicates and over the transcripts of each path), the long-read effect is $\boldsymbol{\Delta}^{\text{L}} = \boldsymbol{n}_{c_1}/\|\boldsymbol{n}_{c_1}\|_1 - \boldsymbol{n}_{c_0}/\|\boldsymbol{n}_{c_0}\|_1$, and the test agrees if $(\boldsymbol{\Delta}^{\text{T}})^{\mathsf{T}}\boldsymbol{\Delta}^{\text{L}} > 0$ for the Tealeaf effect $\boldsymbol{\Delta}^{\text{T}} \in \mathbb{R}^S$. We rank evaluable tests by p-value, then by decreasing statistic. With $a_j \in \{0, 1\}$ the agreement of the test at rank j and $c_r = r^{-1} \sum_{j \le r} a_j$ the cumulative agreement, the endpoint is the normalized area $A_{100} = \frac{1}{100} \sum_{r=1}^{100} c_r$, the mean cumulative agreement over the 100 top-ranked tests. We rank and score the comparators in the same way, with long-read counts assigned by the same annotation and the same use of exon chains. We compute the junction usage of LeafCutter, MAJIQ and scQuint features from the junctions of each counted transcript's exon chain on both strands, and score each event of SUPPA2 and rMATS from transcripts whose exon chain has its inclusion or its exclusion form.

4.11 Comparators

We run the comparators on donor–cell-type pseudobulks. We align the reads to the whole genome with STARsolo (STAR 2.7.11b) [23, 24] and the same annotation as Tealeaf, with the three barcode segments and the UMI of read 2, barcode correction within two edits, UMI collapse with one mismatch, unique alignments only and no cell filter, and sum the junction UMI counts over the cells of each donor and cell type. For MAJIQ and rMATS we write one alignment file per pseudobulk with its unique spliced reads, deduplicated by cell barcode, UMI and alignment. A contrast is a pair of cell types with at least 4 donors observed in both. We run every method on the full data and on each half.

LeafCutter. We cluster introns once over all pseudobulks with LeafCutter [14] (maximum intron length 500 kb, at least 30 reads per cluster, minimum intron ratio 0.001) and test each contrast with `leafcutter_ds` (version 0.2.9), requiring at least 3 samples per intron and per group and 5 reads per cluster in a sample. With LeafCutter's default of 20 reads, about 80 clusters per contrast pass in these pseudobulks, where the median cluster has 2 reads per sample. The donor enters as a covariate, so the test is paired. We take the p-value of each intron cluster and map clusters to genes by their junctions.

scQuint. We group the annotated junctions (both ends at annotated exon boundaries of one gene) by their shared $3'$ splice site, keep groups with at least two junctions, and test each contrast with the Dirichlet-multinomial likelihood-ratio test of scQuint [17], requiring at least 3 samples per intron and per intron group and a global proportion of at least 0.001 per intron. The test has no donor term. A failed fit has $p = 1$.

MAJIQ. MAJIQ v3 [16] builds splice graphs without intron retention from junctions seen in at least 3 samples and quantifies each local splice variant (LSV), the junctions that share a splice site or exon, with at least 3 reads. Heterogen compares the two cell types with the Mann–Whitney test on per-sample posterior means of percent spliced in (PSI), from 100 posterior samples per sample, and requires an LSV to be quantified in half of each group's samples in the full data and in 3 samples in each half. We take the raw Mann–Whitney p-value of each edge (junction) of an LSV.

rMATS. rMATS 4.1.2 [13] counts the junction and exon-body reads (JCEC) of all pseudobulks in one pass (single-end reads of variable length) for annotated skipped exons, alternative $5'$ and $3'$ splice sites, mutually exclusive exons and retained introns. PAIRADISE [30] tests each contrast on the JCEC counts with the two cell types of each donor paired, and we take its p-value without a further filter.

SUPPA2. SUPPA2 2.4 [12] generates events with strict boundaries from the annotation, and an event needs at least one inclusion and one exclusion transcript. We compute event PSI per pseudobulk from spliced transcript abundances (salmon with alevin-fry and parsimony EM, summed over the cells of each pseudobulk) and compare the two cell types of each donor with a paired signed-rank test on PSI differences, on events and contrasts with at least 8 donor pairs. The test is exact, with zero differences dropped and ties handled by the exact conditional distribution. The brain event catalogue contains no alternative first or last exons; the blood catalogue does.

Primer-aware SUPPA2. This variant allocates alevin-fry equivalence-class counts of paired poly(dT) and random-hexamer half-cells to the inclusion and exclusion transcripts of each SUPPA2 event, per donor, cell type and primer. It estimates a shared PSI on the logit scale with an event-specific primer offset, the weighted logit difference between the primers shrunk toward 0 with prior weight 200, and a Beta-binomial overdispersion estimated by moments and shrunk toward 0.05 with prior weight 200, separately in the full data and in each half. The test is the exact paired signed-rank test of SUPPA2.

Aggregation. We map every feature to a gene, drop features without one, and combine feature p-values into (gene, cell-type pair) keys and genes by Simes' method, as for Tealeaf blocks. A method leaves out a contrast in which no feature passes its filters. In the brain full data MAJIQ tests 170 of the 190 contrasts and the other methods all 190; in GSE320507 MAJIQ tests 155 of the 171 contrasts, scQuint 170 and the others all 171.

4.12 Coverage-only null

We recompute each method's tests on counts with the coverage of the data and no difference between the two cell types. For a test of a contrast, every sample (a subject–cell-type row for Tealeaf, a pseudobulk for the comparators) keeps its total count on the tested feature, and we redraw the counts of the feature's components from proportions pooled over the samples of both cell types. For Tealeaf the components are the compatibility classes of the block, which we redraw with each row's molecule total per primer from one set of path shares without subject or row variation, and we refit the test with the model and settings used on the data (a failed fit has $p = 1$). For LeafCutter and scQuint we redraw the junction counts of an intron cluster or intron group from a multinomial distribution and run the method with the settings used on the data on the redrawn features. For rMATS we redraw the inclusion count $I_s \in \mathbb{N}_0$ of an event in sample s as Binomial$(I_s + S_s, f_0)$, with $S_s \in \mathbb{N}_0$ the skipping count and $f_0 = \sum_s I_s / \sum_s (I_s + S_s)$, and rerun PAIRADISE. Supplementary Note 2 describes the null for MAJIQ. In each run of a method (on the full data and on each half), we recompute the tests with $p < 0.01$ on the data on three draws and remove a test when the median of its three null p-values is below 0.05. We keep tests with $p \geq 0.01$. We do not recompute SUPPA2 and primer-aware SUPPA2; they replicate 0 to 8 genes, so removal would not change the comparison. For a test with null p-values p_1^0, p_2^0, p_3^0, we remove the test when median$(p_1^0, p_2^0, p_3^0) < 0.05$, which for a calibrated test removes $3(0.05)^2(0.95) + (0.05)^3 \approx 0.7\%$ of the recomputed tests at random. We apply removal in every run separately, the full data for long-read direction and discoveries and each half for split replication, and form the universes after removal.

4.13 Human blood data

We processed three public Parse datasets of human peripheral blood mononuclear cells (PBMCs) by the same pipeline as the brain data, with the GRCh38 genome and the GENCODE v49 basic annotation, all multi-exon genes as targets, two alignment passes and novel paths. We remove a leading template-switch oligo from read 1 before alignment, since on short reads the remainder would fall below the minimum aligned length.

1. **GSE320507** [32], PBMCs of donors with long-term COVID-19 syndrome (long COVID, LTCS), convalescent donors (CONV) and healthy controls (HC). Each sample occupies one first-round well. The well-to-donor map provided by the authors agrees with genotypes called from expressed variants in every well but one, which holds cells of two donors and which we exclude. This leaves 38 donors (26 LTCS, 5 CONV, 7 HC).

2. **GSE327907** [33], PBMCs of 12 donors in two batches under 12 in-vitro conditions; only the unstimulated cells enter the cell-type contrasts.

3. **GSE334840** [34], PBMCs of 8 donors in two hormone treatments and a vehicle control (DMSO); only the control cells enter the cell-type contrasts.

Cell types are CellTypist [35] labels with majority voting, from the fine-grained immune model for GSE320507 (19 types) and from the coarse model for the stimulation series, with macrophages merged into monocytes (10 and 8 types), because fine-grained labels separate stimulated monocytes by stimulus. We test every pair of cell types and apply BH over the internal and terminal tests of the cell-type pairs of a cohort. We split donors into halves stratified by group (GSE320507) or batch (GSE327907). For Tealeaf alone, without a comparator, the split universe is the set of (gene, cell-type pair) keys with a Tealeaf test in both halves, and we score replication as in Section 4.10; in GSE320507 we also score Tealeaf against each comparator on the comparator's universes, as in the brain data.

Long-read directions come from a public PacBio Kinnex library made from a Parse whole-transcriptome library of commercial PBMCs, from donors other than those of the three datasets. The library provides segmented reads without cell calls, so we call the three barcodes, the UMI and the reverse-transcription primer of each read. We place the two linkers by semi-global alignment, correct each barcode to the unique barcode within edit distance 1, and trim the poly(T) and the template-switch oligo. Of 148 million reads, 93% are assigned to a cell. We align reads to the genome with minimap2 [36] in spliced mode with the annotated junctions as hints. We assign a read with mapping quality at least 5 to the gene with which it has the largest same-strand exonic overlap, and reads of a cell and gene whose UMIs differ by at most one base form a molecule, which takes the most frequent exon chain of its reads. We label cells with at least 1,000 reads (10,077 cells, 26.4 million molecules) by CellTypist with the model used for each short-read dataset.

In GSE320507 we compare the disease groups within each cell type, with the levels in different donors (Section 4.8) and at least 4 donors in each level. Each cell type has four contrasts, LTCS against HC, LTCS against CONV, CONV against HC, and LTCS against HC and CONV pooled. A fifth contrast compares two sets of 26 and 12 donors drawn at random from the 38 without regard to group, the same draw in every cell type. Discoveries are BH-adjusted $p \leq 0.05$ within each contrast over its internal and terminal tests, with tests that do not converge entering at $p = 1$. In the long-read data, we use the cell types with at least one evaluable test and score directions as in Section 4.10.

4.14 Event classes and discoveries

We assign each test an event class from the exon chains of its retained paths. If all chains start and end at the same bases the block is internal, and we classify two chains by the exons in which they differ as a cassette exon, multi-exon skipping, an alternative donor or acceptor, mutually exclusive exons or a retained intron (a differing pattern outside these is complex internal); more than two chains are compound internal. Otherwise the paths differ at a transcript end, an alternative first exon, an alternative last exon, or both, oriented by the strand; when the differing terminal exons share their splice site, only the transcript start or end moves. For discoveries, each method's finite p-values on the contrasts it shares with every other method form one BH family, and a discovery has adjusted $p \leq 0.05$. Methods test different units, a block for Tealeaf, an intron cluster for LeafCutter, an intron group for scQuint, an LSV for MAJIQ and an event for SUPPA2 and rMATS, so we also count discoveries as gene–cell-type pairs with at least one discovery, which take the class of their smallest-p discovery. A MAJIQ LSV takes the smallest p-value of its edges that remain after the

coverage-only filter. We map SUPPA2 and rMATS events to the Tealeaf classes, skipped exon to cassette exon, alternative 5′ and 3′ splice sites to alternative donor and acceptor, and alternative first and last exons, mutually exclusive exons and retained introns to themselves.

4.15 Examples

Brain examples are the full-data tests with the largest effect among those evaluable in the long-read data, selected without regard to long-read agreement. Candidates have $p < 10^{-8}$, at least 50 long-read UMIs per cell type and at most four paths; we take one per gene, rank by the norm of the fitted path-usage difference and require at least one internal and one terminal block among the examples. Disease-contrast examples are LTCS against HC discoveries with at most four paths, a median of at least 50 gene molecules per donor row and a block window of at most 20 kb, one per gene, ranked by the same norm. The window limit keeps the block view readable; without it the first choice is a terminal block of *RNF213* spanning 79 kb and 52 exons. The path usage of each row is the molecule-share composition estimated by the EM algorithm of the share screen with primers pooled. We draw sashimi plots with ggsashimi [37] from UMI-deduplicated alignments of the targeted aligner.

4.16 Implementation

The alignment kernel has two implementations with identical output, a numba implementation in the Python package and a Rust library called through a C interface. The Rust library also reads the paired gzipped FASTQ files. It decompresses and parses each mate on its own thread ahead of alignment, converts read 1 to 2-bit codes and reduces read 2 to the corrected barcode and UMI. A run of 345 million read pairs aligns to the 1,163 genes of an earlier target list in 45 min with 16 threads, and to all 33,755 multi-exon genes in 32–67 min with 18 threads, compared with 67 min for STARsolo on 12 threads over the whole genome. Read opportunities, the row integrals and the subject quadrature are compiled numba kernels. Against STARsolo alignments of 2 million reads (earlier target list), the targeted aligner reports 97.6% of STAR's unique alignments in the target loci as unique, with identical junctions for 99.5% of reads aligned uniquely by both, and gives the same block class for 99.7% of reads entering a block. Molecule counts per block, class, subject, cell type and primer agree with those from STARsolo alignments to within 5% (overlap of 0.95 of STAR's counts), and the class distributions within blocks differ by a mean total-variation distance of 0.005.

Code and data availability

Tealeaf, the aligner and the evaluation scripts are available at the project repository. GSE320507, GSE327907 and GSE334840 are public in the Gene Expression Omnibus, the Kinnex PBMC library is a public PacBio dataset, and the mouse brain long-read data are those of Joglekar et al. [29].

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Supplementary Information

Supplementary Figures

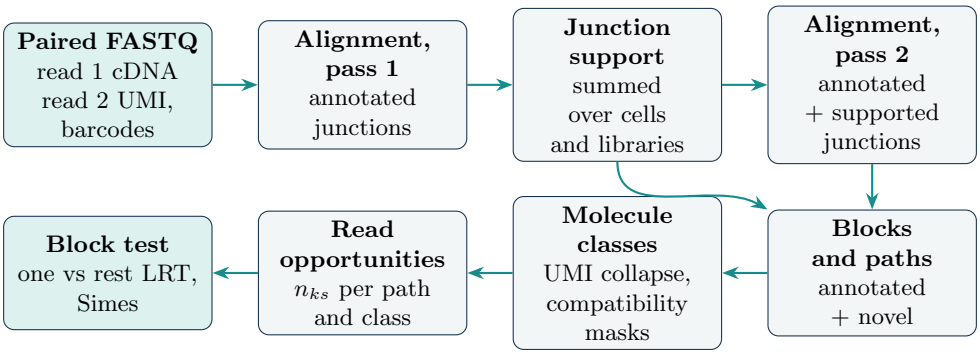


Figure S1: Tealeaf from reads to block tests. Junction support from the first alignment pass defines the known junctions of the second pass; support from the second pass defines the novel paths. No step before the test uses cell-type labels.

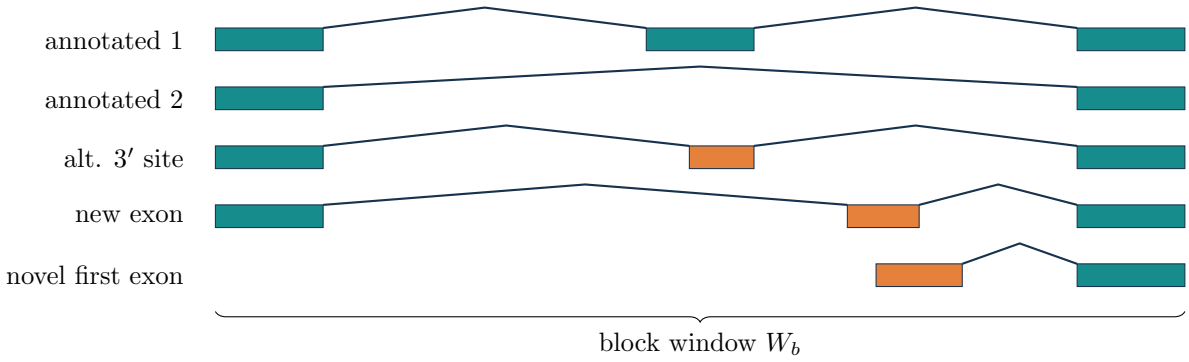


Figure S2: Novel paths in a block with two annotated paths (teal). Orange exons are created from novel junctions, an alternative acceptor in an existing exon, a new exon from a pair of novel junctions, and a novel first exon of length T from a donor deep in an intron (more than 300 bases from any exon) whose junction pairs with no other.

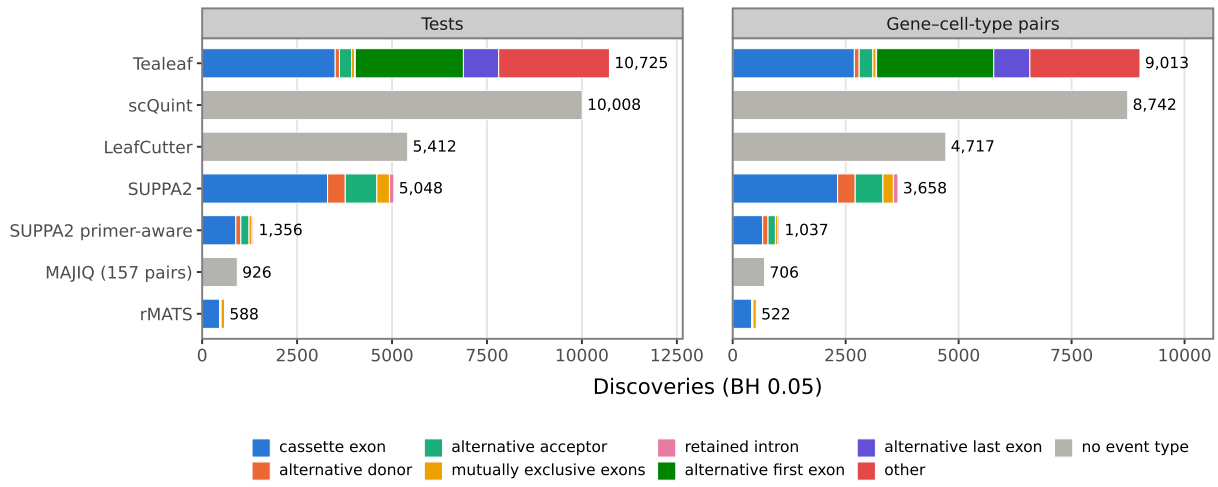


Figure S3: Full-data discoveries as in Fig. 2d, on all 190 cell-type pairs each method tests; methods tested on fewer pairs give their number of pairs.

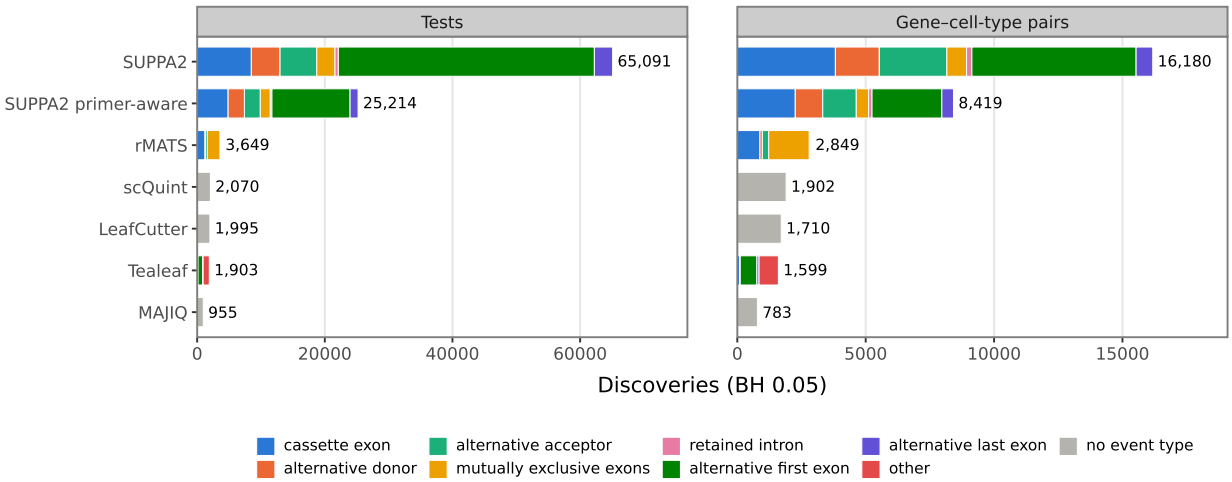


Figure S4: GSE320507 full-data discoveries (BH 0.05 within method) on the 144 cell-type pairs tested by every method, counted as tests (left) and as gene–cell-type pairs with at least one discovery (right), by event class, as in Fig. 2d. Tests called by the coverage-only null are removed for every method except SUPPA2.

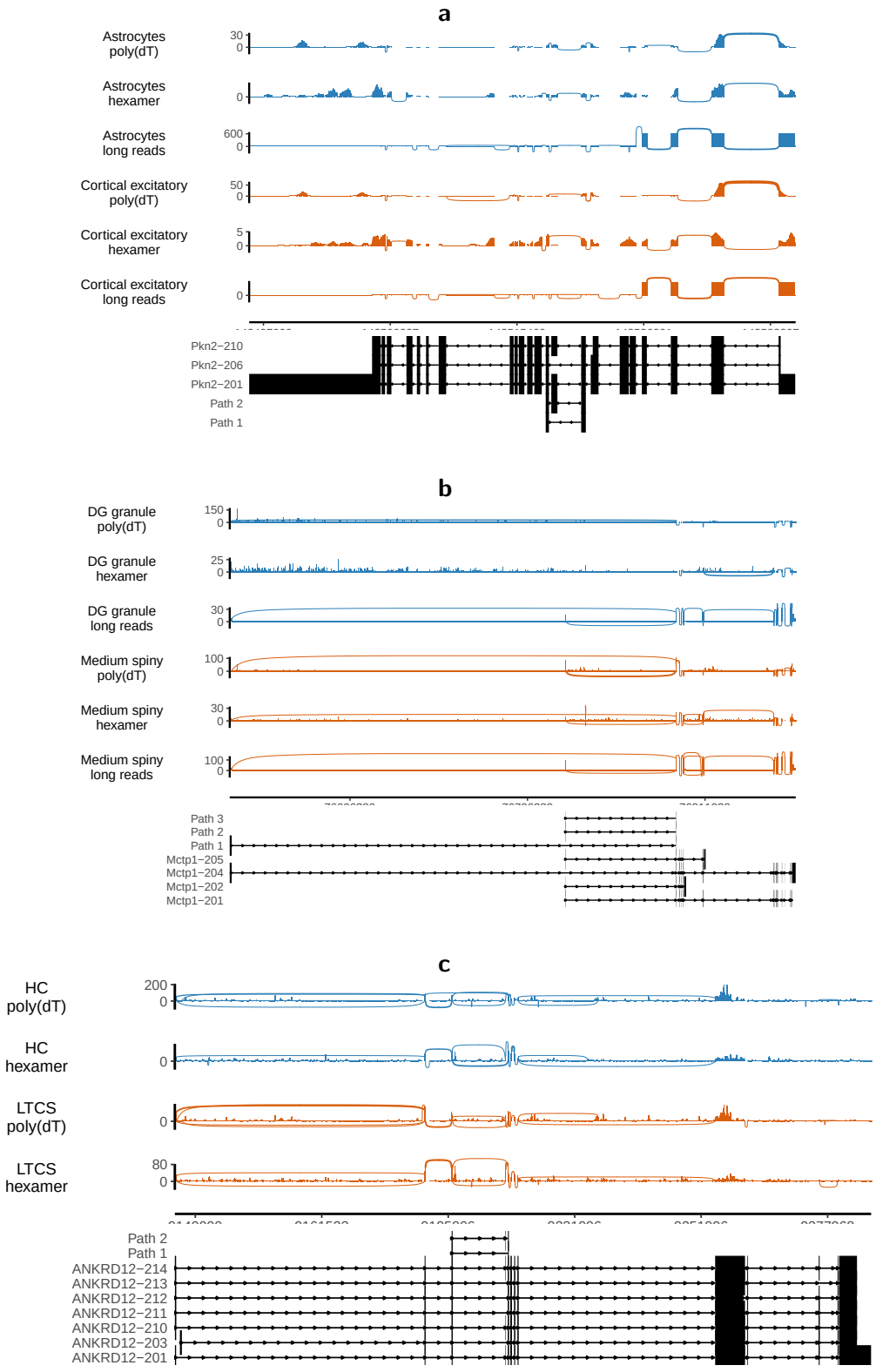
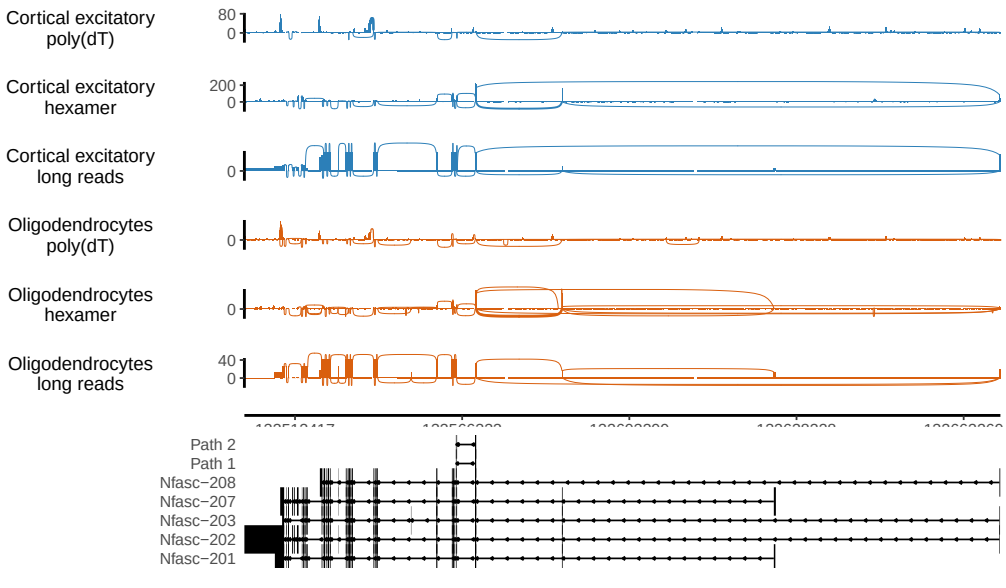
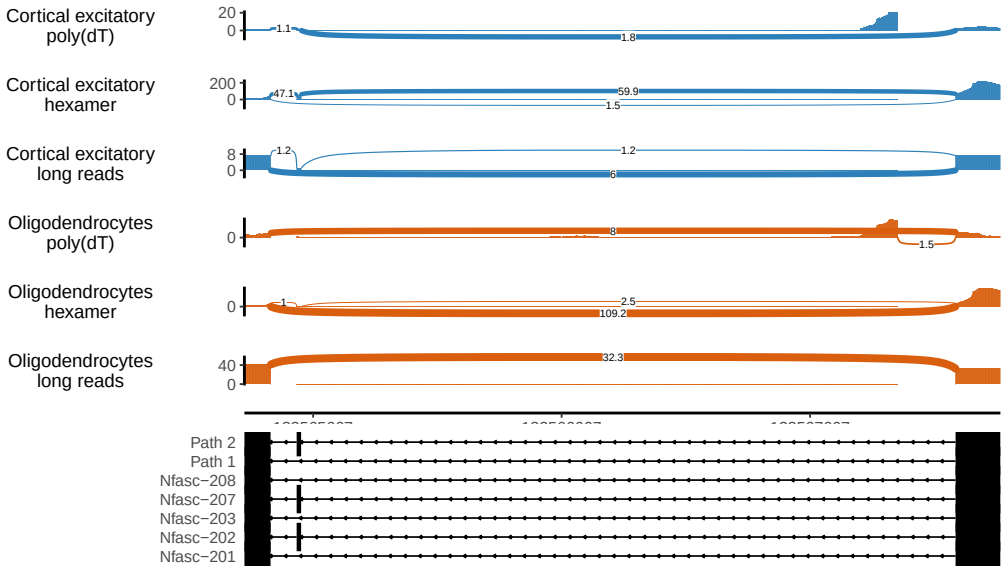


Figure S5: Whole-gene sashimi views of the examples of Figs. 4 and 5: **a**, *Pkn2*; **b**, *Mctp1*; **c**, *ANKRD12*. Tracks as in the block views; junction labels are omitted, and the block's paths mark the block.

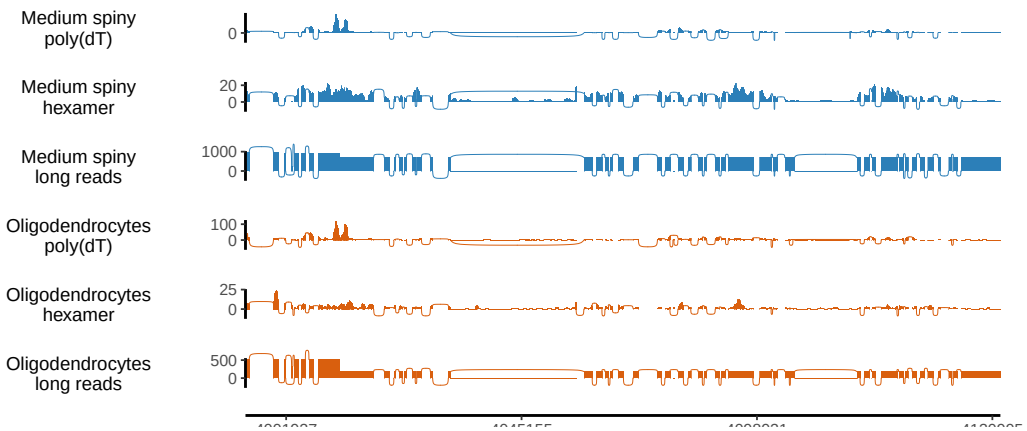


(a)

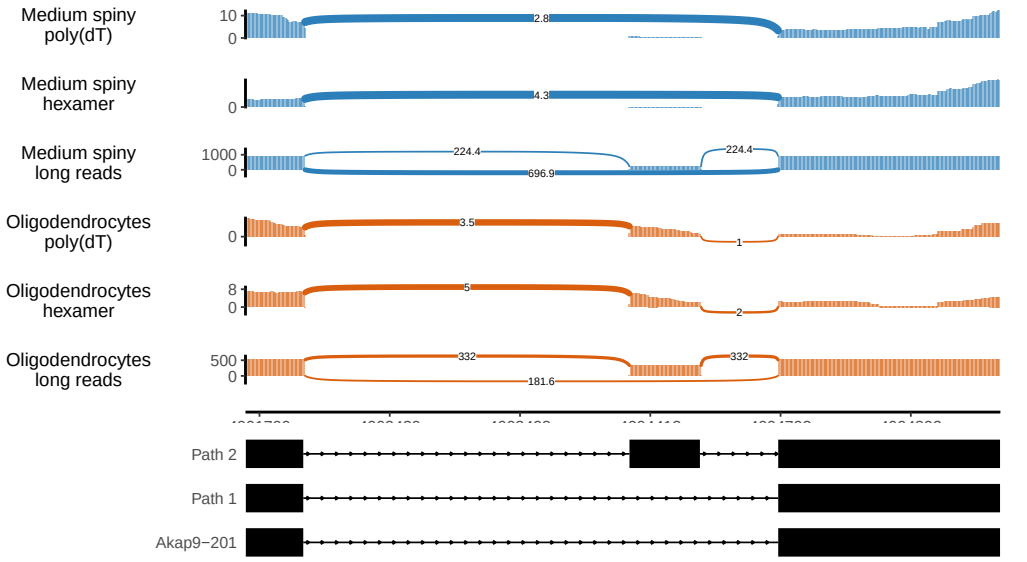


(b)

Figure S6: Nfasc block 2, cortical excitatory neurons against oligodendrocytes. Panels as in Fig. 4a, with the whole gene in (a).



(a)



(b)

Figure S7: *Akap9* block N2, a block built from the data around a 54-base cassette exon that the basic annotation lacks, in medium spiny neurons and oligodendrocytes. Panels as in Fig. 4a, with the whole gene in (a).

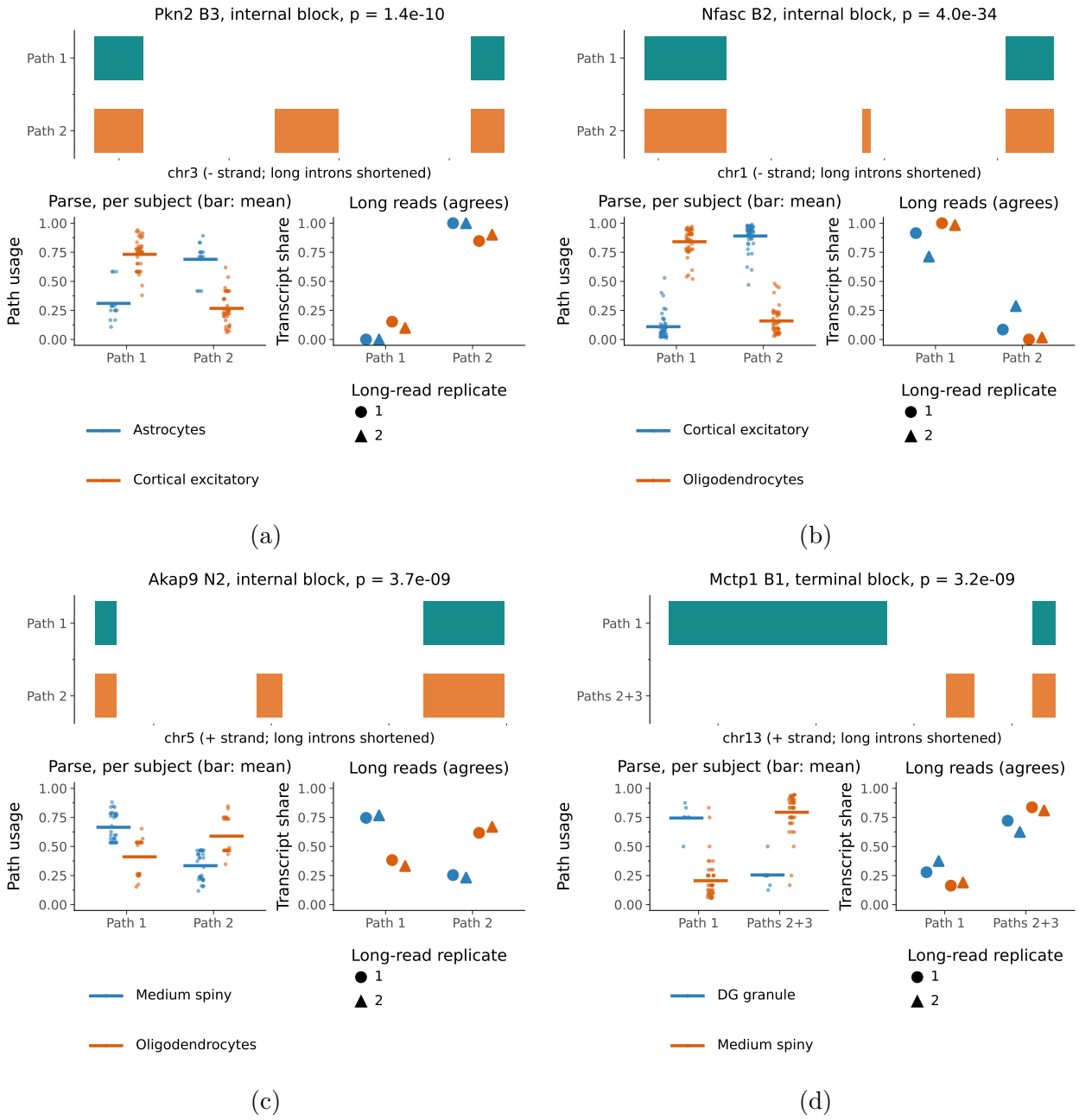
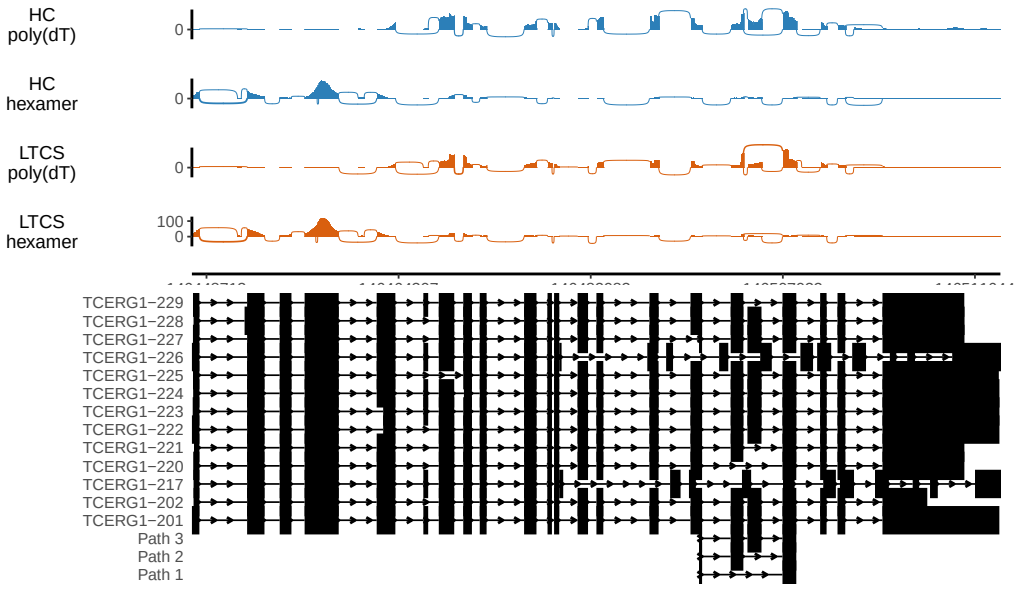
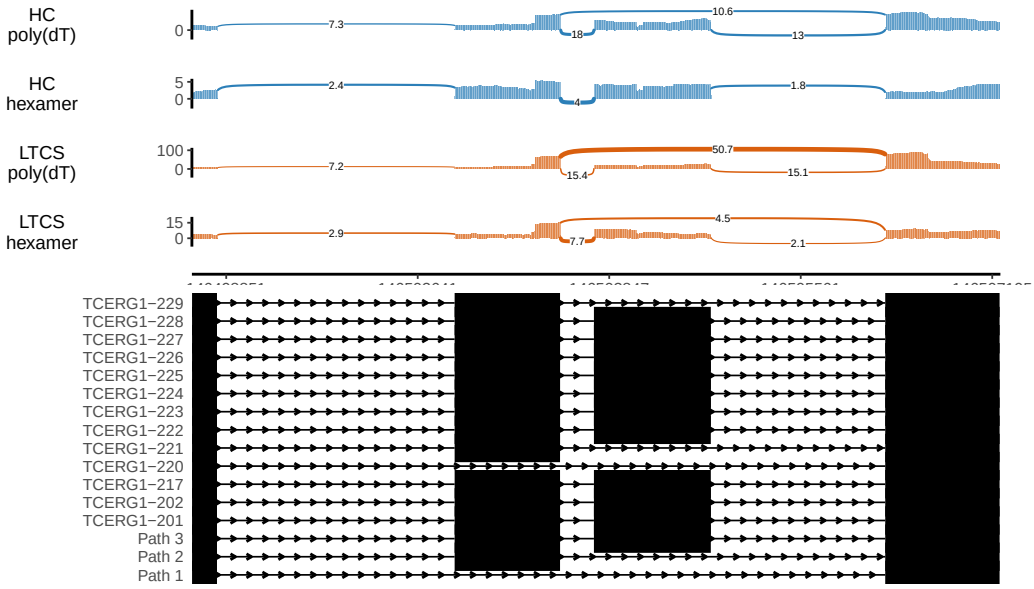


Figure S8: Path usage in the four examples: (a) Pkn2, (b) Nfasc, (c) Akap9, (d) Mctp1. In each panel, top, the tested paths, numbered as in the sashimi plots (introns longer than 300 bases shortened; merged paths are drawn by their first member); bottom left, the path usage of each subject in the Parse data with the mean over subjects (bars); bottom right, the transcript share of each path in the two long-read replicates.

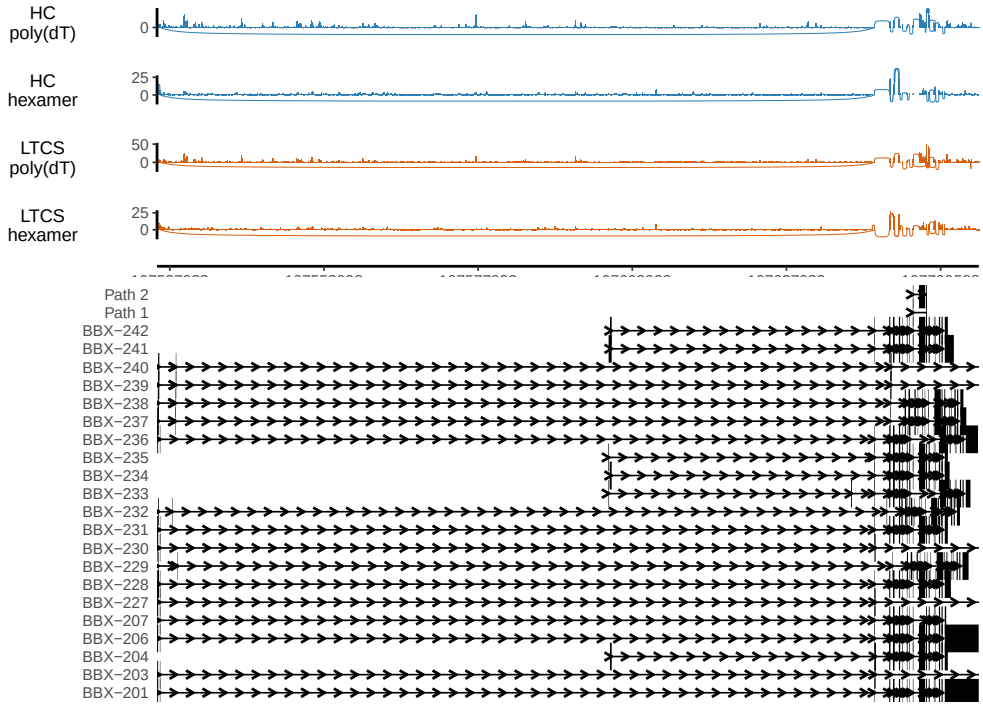


(a)

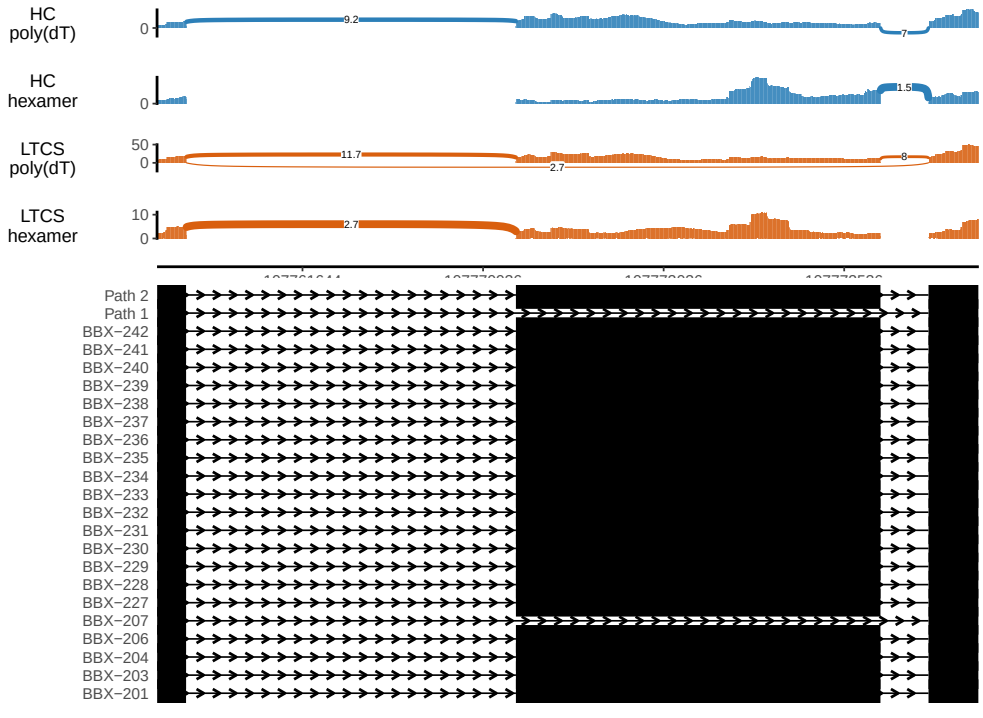


(b)

Figure S9: TCERG1 block 7 in naive and central-memory helper T cells, HC against LTCS donors. Panels as in Fig. 5e, with the whole gene in (a).

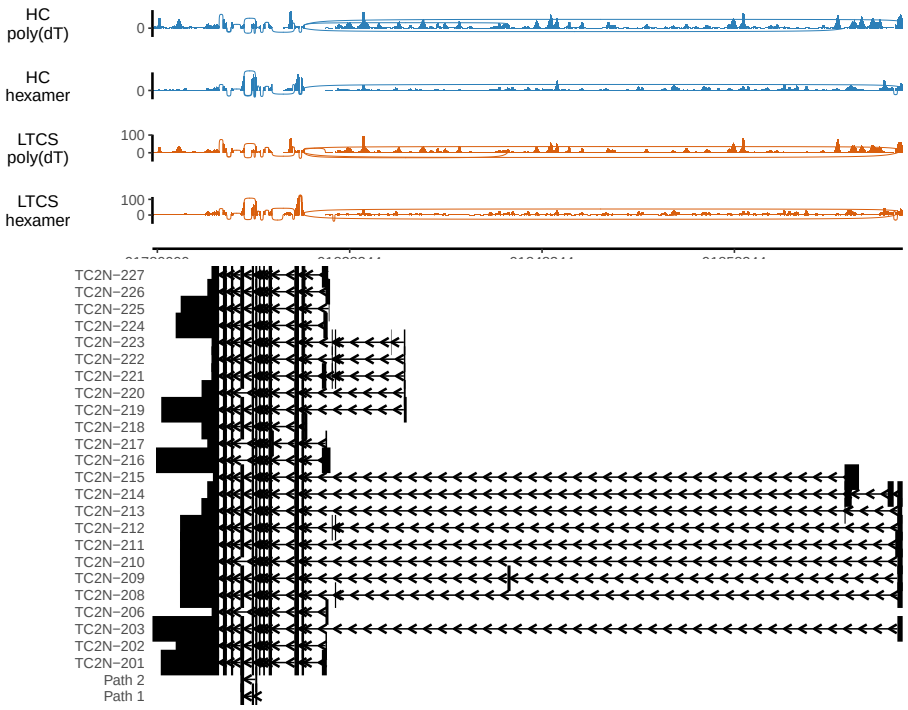


(a)

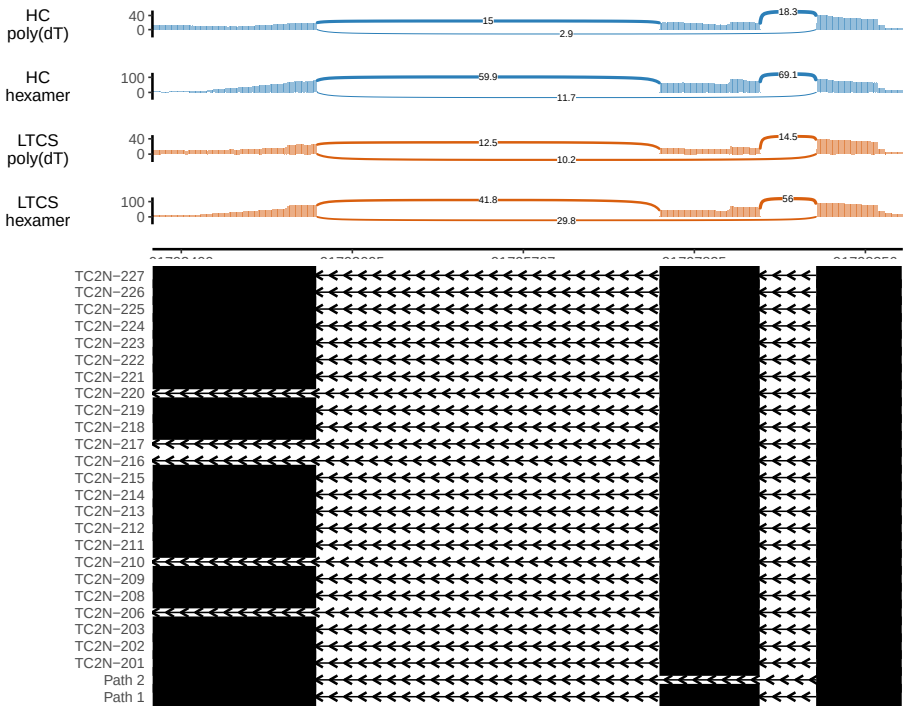


(b)

Figure S10: BBX block 4 in naive and central-memory helper T cells, HC against LTCS donors. Panels as in Fig. 5e, with the whole gene in (a).



(a)



(b)

Figure S11: TC2N novel block 3 in naive and central-memory helper T cells, HC against LTCS donors. Panels as in Fig. 5e, with the whole gene in (a).

Supplementary Tables

	LeafCutter	MAJIQ	SUPPA2	SUPPA2-PA	rMATS	scQuint
Matched universe						
Genes	1,509	2,758	3,411	3,072	5,070	3,603
Tealeaf replicated	182	205	130	138	167	192
Comparator replicated	165	17	8	0	4	134
Comparator universe						
Genes	1,561	2,996	5,068	3,901	6,838	4,220
Tealeaf replicated	182	196	105	124	139	170
Comparator replicated	169	15	0	0	4	123
Comparator A_{100}	.936	.944	.797	.622	.898	.912

Table S1: Replicated genes on each comparator's matched and comparator universes, and long-read direction. SUPPA2-PA is primer-aware SUPPA2. Tests called by the coverage-only null are removed for Tealeaf and every comparator except SUPPA2 and SUPPA2-PA (Section 4.12; Supplementary Note 4 gives the results without removal). Tealeaf has $A_{100} = 0.947$ (95 of its 100 top-ranked evaluable tests agree with long reads; $A_{40} = 0.965$, $A_{200} = 0.943$).

Event class	Tests	Discoveries	Novel path	LR agreement	Top 100
cassette exon	217,758	3,549	1,553	0.89 (456)	35/36
alternative first exon	101,666	2,872	2,234	0.90 (146)	23/23
compound internal	104,884	1,642	1,252	0.84 (75)	1/1
alternative last exon	57,878	934	644	0.85 (74)	21/25
alternative first and last exons	24,646	920	776	0.83 (48)	9/9
alternative acceptor	100,863	323	259	0.79 (38)	2/2
alternative transcript end	9,055	156	38	1.00 (11)	1/1
alternative donor	57,294	114	92	0.50 (12)	0/0
multi-exon skipping	8,988	105	56	0.91 (22)	3/3
mutually exclusive exons	2,701	86	12	0.82 (11)	0/0
complex internal	4,982	59	57	1.00 (3)	0/0
alternative transcript start	2,994	57	8	1.00 (2)	0/0
retained intron	1,244	13	13	– (0)	0/0

Table S2: Full-data tests by event class. Novel path counts discoveries that retain an unannotated path. LR agreement (long-read agreement) is the fraction of long-read-evaluable discoveries whose direction agrees with the long reads (number evaluable in parentheses); Top 100 gives agreeing over all tests of the class among the 100 top-ranked long-read-evaluable tests.

	GSE320507	GSE327907	GSE334840
Donors (per half)	38 (19)	12 (6)	8 (4)
Cells	70,643	24,314	23,000
Cell types	19	10	8
Read 1 length	136	150	66
Read pairs (billions)	1.84	11.7	2.52
Full-data tests	1,504,546	200,346	80,470
Discoveries	2,340	472	15
Gene–pair discoveries	1,847	377	13
Genes tested in both halves	10,079	8,640	5,211
Discoveries, half 1 / half 2	133 / 96	15 / 29	0 / 0
Replicated genes	65	5	0

Table S3: Cell-type contrasts in three PBMC datasets. Cells are those of the contrasts (unstimulated or control cells for GSE327907 and GSE334840); read pairs are for all cells of the dataset. Discoveries are tests with BH-adjusted $p \leq 0.05$, and gene–pair discoveries (gene, cell-type pair) keys with Simes-combined adjusted $p \leq 0.05$. Half discoveries and replicated genes are gene-level, on the genes tested in both halves.

	GSE320507	GSE327907	GSE334840
Long-read cell types (pairs)	14 (91)	7 (20)	7 (20)
Evaluable tests	44,359	17,488	11,932
Agreeing, top 40	39	38	35
Agreeing, top 100	99	96	82
Agreeing, top 200	197	177	152
A_{100}	0.988	0.967	0.865
Evaluable discoveries	497	138	6
Agreement of discoveries	0.88	0.94	1.00
Agreement of all evaluable tests	0.54	0.54	0.52

Table S4: Long-read direction of full-data cell-type tests in three PBMC datasets, against molecules of a Parse PBMC long-read library. Long-read cell types and pairs are those with at least one evaluable test. Agreeing gives the number of the r top-ranked evaluable tests whose direction agrees with the long reads.

	LeafCutter	MAJIQ	SUPPA2	SUPPA2-PA	rMATS	scQuint
Matched universe						
Genes	3,461	3,411	8,478	8,405	8,829	6,034
Tealeaf replicated	67	69	57	57	65	55
Comparator replicated	120	23	0	0	131	66
Comparator universe						
Genes	3,588	3,618	10,577	10,569	10,179	6,402
Tealeaf replicated	67	68	56	53	60	54
Comparator replicated	126	26	0	0	133	67
Comparator A_{100}	.991	1.000	.555	.460	.575	1.000

Table S5: GSE320507 cell-type contrasts: replicated genes on each comparator's matched and comparator universes, and long-read direction against the Kinnex library, as in Supplementary Table S1. Tests called by the coverage-only null are removed for Tealeaf and every comparator except SUPPA2 and SUPPA2-PA. Tealeaf has $A_{100} = 0.988$ (99 of its 100 top-ranked evaluable tests agree; $A_{40} = 0.992$, $A_{200} = 0.989$).

Contrast	Tests	$p < 0.01$	$p < 0.001$	Discoveries
Random 26 vs 12	182,434	0.44%	0.041%	0
CONV vs HC	62,155	0.70%	0.069%	0
LTCS vs CONV	66,481	0.57%	0.032%	0
LTCS vs HC	148,293	1.65%	0.460%	215
LTCS vs HC and CONV	182,408	1.18%	0.251%	23

Table S6: Disease-group contrasts within cell types in GSE320507, full data. Tests that do not converge enter with $p = 1$; discoveries are BH-adjusted $p \leq 0.05$ within each contrast.

Method	Full data	Half 1	Half 2
Tealeaf	147 / 24,176	65 / 10,540	50 / 9,687
LeafCutter	45 / 8,648	28 / 4,445	26 / 3,513
scQuint	248 / 29,391	110 / 13,606	103 / 13,129
rMATS	1,435 / 17,114	1,598 / 12,836	1,126 / 10,738
MAJIQ	4,199 / 6,158	5,932 / 13,765	13,625 / 21,434

Table S7: Brain tests removed by the coverage-only null over tests recomputed ($p < 0.01$ on the data). MAJIQ counts are connections (edges) of local splice variants.

Supplementary Notes

1 Differential intron retention

Block tests do not test retention of a whole intron, since a path whose chain is the block window duplicates the precursor path and is removed (1,244 retained-intron tests remain, Table S2). We therefore test retention per annotated intron. In nuclei most intronic molecules are precursor RNA, so the retained fraction of an intron is compared with that of the other introns of its gene.

1.1 Counts

Introns are the junctions of the comprehensive GENCODE vM32 transcripts of the target genes. An intron is *clean* if no exon boundary of any transcript, of any gene and on either strand, lies strictly inside it; clean introns are disjoint. For clean intron i, subject u and cell type c, $r_{iuc} \in \mathbb{N}_0$ is the number of retained molecules, sense molecules with an unspliced read inside the intron at least 5 bases from both ends, and $n_{iuc} \in \mathbb{N}_0$ the number of spliced molecules, those with a junction equal to the intron. Molecules are distinct half-cell barcode and UMI pairs, and both primers are pooled. Poly(dT) priming at genomic A-runs inflates intronic coverage, intron-body poly(dT) molecules per position being 27.5-fold higher within 100 bases upstream of a sense A-run (at least 8 A in 10 bases) than more than 3 kb away, 3.4-fold at 200–300 bases and 2- to 2.7-fold up to 3 kb, against 1.2- to 1.9-fold for random-hexamer molecules. We remove poly(dT) intron-body molecules within 1 kb upstream of a sense A-run. The gene baseline of row (u, c) is $R_{iuc} = \sum_{j \neq i} r_{juc}$ and $N_{iuc} = \sum_{j \neq i} n_{juc}$, with j over the other clean introns of the gene.

1.2 Model and test

A test is indexed by a clean intron i and a cell-type pair $\{c_0, c_1\}$, with rows (u, c), $c \in \{c_0, c_1\}$, that have $r_{iuc} + n_{iuc} \geq 3$ and $R_{iuc} + N_{iuc} \geq 3$; at least 4 subjects must have rows in both cell types. Dropping the index i, the model is the random-effects model of the block test (Section 4.8) with one path against the rest and exact rows,

$$r_{uc} \mid \psi_{uc} \sim \text{Binomial}(r_{uc} + n_{uc}, \psi_{uc}), \qquad \psi_{uc} \mid v_u \sim \text{Beta}\left(\kappa\mu_{uc}, \kappa(1-\mu_{uc})\right), \qquad v_u \sim \mathcal{N}(0, \sigma^2),$$

with

$$\text{logit}\, \mu_{uc} = \theta_0 + \theta_1 z_c + v_u + o_{uc}, \qquad o_{uc} = \log \frac{R_{uc} + 0.5}{N_{uc} + 0.5},$$

where $\psi_{uc} \in (0, 1)$ is the retained fraction of the row, $z_c = \mathbf{1}[c = c_1]$ and the offset $o_{uc} \in \mathbb{R}$ is fixed. θ_1 is then the change in the intron's retention relative to its gene, and a change of the whole gene's unspliced fraction between cell types enters through o_{uc}. Integration, estimation, bounds and the convergence check are those of the block test, and the likelihood-ratio statistic T compares fits with and without θ_1. We call this test the Beta-binomial test.

Calibration. In simulations of this model with k subjects, $P(\chi^2_1 > T)$ rejects 10–12% of null tests at 0.05 for $k \leq 6$, and the signed root of T on t_{k-1} is conservative. On the data, we generate the null by refitting 15% of the tests, chosen at random, with each subject's two cell-type labels exchanged with probability 1/2. This null rejects 8–10% at 0.05, 2–3% at 0.01 and 0.2–0.6% at 0.001 under χ^2_1, at every number of subjects from 4 to more than 18. 15–27% of null statistics are below 10^{-3}, and above their 80th percentile they follow χ^2_1 scaled by a constant. With q_α the upper-α quantile of the null statistics and $\chi^2_{1,\alpha}$ that of χ^2_1, the scale is $\varrho = \frac{1}{3} \sum_{\alpha \in \{0.05, 0.01, 0.001\}} q_\alpha / \chi^2_{1,\alpha}$, estimated in each cohort, and the p-value is $P(\chi^2_1 > T/\varrho)$. ϱ is 1.30 and 1.34 in the two halves and 1.34 in the

full data. A scaled t distribution fitted by maximum likelihood to the null signed roots matches the bulk of the null but has a heavier tail and leaves no discoveries.

Per-subject comparison. As a simpler alternative, $x_{uc} = \log\{(r_{uc}+0.5)/(n_{uc}+0.5)\} - o_{uc}$ with variance $w_{uc} = \sum_{y\in\{r,n,R,N\}}(y_{uc}+0.5)^{-1}$ gives per-subject differences $d_u = x_{uc_1} - x_{uc_0}$ with variance $w_{uc_0} + w_{uc_1}$, combined over the k subjects with both cell types by the DerSimonian–Laird random-effects mean, with the Hartung–Knapp variance (floored at the DerSimonian–Laird variance) and t_{k-1}.

1.3 Evaluation

We leave cerebellar granule neurons out of the evaluation. Relative to the median of the eight long-read cell types, their gene-wide log ratio of intron-body to spliced molecules is -1.99 with poly(dT) and -2.07 with random hexamers (astrocytes -0.43 and -0.49, other types within ± 0.13). At introns without retained short-read molecules a test with the gene offset takes the sign of this baseline difference, and in the 28 pairs these tests make up most of the top-ranked long-read-evaluable tests of the per-subject comparison. The evaluation uses the 21 pairs of the other seven long-read cell types. We score split replication as in Methods, Section 4.10, with intron p-values combined by Simes within a gene–cell-type pair and the pairs of a gene by Simes, on each method's own (gene, pair) keys. In the long-read data a molecule retains a clean intron when one of its exons contains the intron with 20 bases on each side, and splices it when one of its junctions equals the intron. The long-read effect is the difference of retained fractions between the cell types, and a test is evaluable if each cell type has at least 20 long-read molecules retaining or splicing the intron and either (i) at least 5 molecules in the two cell types retain it or (ii) the retained fractions differ by at least 0.05. Agreement is the fraction of the 100 top-ranked evaluable tests whose effect has the sign of the long-read effect. We map SUPPA2 retained-intron events RI:chr:s_1:e_1-s_2:e_2 (1-based) to the clean intron $[e_1, s_2 - 1)$, and MAJIQ, built with intron retention, to its intron edges; an intron in several local splice variants keeps its smallest p-value.

Method	Genes	BH genes per half	Replicated	LR retained ≥ 5	LR difference ≥ 0.05
Beta-binomial, gene offset	4,626	30, 26	10	.61 (5,820)	.58 (516)
pooled retained ≥ 50	4,146	34, 31	10	.65 (637)	– (95)
rows by primer	4,134	24, 14	8	.54 (4,921)	.54 (423)
random hexamer only	3,813	15, 16	5	.54 (2,279)	.57 (285)
Per-subject comparison	4,626	7, 9	2	.45 (5,820)	.81 (516)
random hexamer only	3,813	2, 2	0	.56 (2,279)	.76 (285)
SUPPA2	198	0, 0	0	.43 (657)	.57 (224)
Primer-aware SUPPA2	119	0, 1	0	.45 (494)	.56 (184)
MAJIQ	31	8, 20	6	– (69)	– (22)

Table S8: Differential intron retention on the 21 long-read cell-type pairs without cerebellar granule neurons. Genes is the number of genes with a test in both halves, BH genes per half the genes with BH-adjusted Simes p at most 0.05 in each half, and Replicated the genes with BH-adjusted conjunction p at most 0.05. The last two columns give the fraction of the 100 top-ranked long-read-evaluable tests that agree with the long-read direction under the two evaluability rules (number evaluable in parentheses; – when fewer than 100). The pooled retained filter keeps tests with at least 50 retained molecules over both cell types, without labels. Rows by primer split each row by primer, each with that primer's gene offset.

Table S8 gives the results. The Beta-binomial test replicates 10 of 4,626 genes against 2 for the per-subject comparison, 0 for both forms of SUPPA2 and 6 of 31 for MAJIQ, which tests few introns.

A label-blind filter on pooled retained molecules leaves replication unchanged, rows by primer lower ϱ to 1.23–1.27 but leave fewer tests, and random-hexamer molecules alone halve replication. Long-read agreement of the top-ranked tests is between 0.54 and 0.65 for every Beta-binomial variant and does not favour one test, since the Beta-binomial leads under the retained-molecule rule and the per-subject comparison under the difference rule. Among tests with at least 10 retained short-read molecules and a long-read difference of at least 0.05 (201 tests), the pooled short-read changes in retained fraction, with or without the gene offset, agree with the long-read direction in 51–52% of tests. With these data differential intron retention between cell types is underpowered, with few replicated genes and weak agreement between short-read and long-read retention changes of single introns.

2 MAJIQ Heterogen under the coverage-only null

MAJIQ Heterogen compares two groups of samples with a Mann–Whitney test on per-sample posterior means of PSI. For connection j of an LSV with J connections, sample s with $x_{sj} \in \mathbb{N}_0$ reads on the connection and $n_s \in \mathbb{N}_0$ reads on the LSV has posterior mean $\hat{\psi}_{sj} = (x_{sj}+1/J)/(n_s+1)$. The prior adds one read spread over the J connections, so at low coverage $\hat{\psi}_{sj}$ moves toward $1/J$ and takes few distinct values. When a gene is expressed at different levels in two cell types, the two groups of $\hat{\psi}_{sj}$ then differ without a difference in PSI. Because expression differences between cell types are shared by all subjects, such tests also replicate across subject halves.

The coverage-only null keeps each sample's LSV total n_s and MAJIQ's per-sample quantifiability, draws $x^0_{sj} \sim \text{Binomial}(n_s, p_{0j})$ with $p_{0j} = \sum_s x_{sj} / \sum_s n_s$ pooled over the samples of both groups, and recomputes the test on $(x^0_{sj}+1/J)/(n_s+1)$. We reproduce the test from the per-sample totals and connection reads in MAJIQ's coverage files, with the minimum number of samples per group of each run, as an asymptotic Mann–Whitney test with tie correction; its Spearman correlation with MAJIQ's p-values is 0.995 or more in every run (MAJIQ's p-values are smaller at complete separation, where it uses the exact distribution). We make three null draws per test and remove an edge with MAJIQ $p < 0.01$ when the median of its three null p-values is below 0.05. For long-read direction and discoveries, an LSV takes the smallest MAJIQ p-value over its remaining edges and is dropped when no edge remains.

Run	Tests	Reproduced test	Coverage-only null
Brain, full data	26,353	6,542	4,816
Brain, half 1	518,233	1,383	880
Brain, half 2	438,421	7,859	6,094
GSE320507, full data	304,221	121,768	119,007
GSE320507, half 1	359,620	111,173	110,246
GSE320507, half 2	311,355	82,037	82,260

Table S9: MAJIQ Heterogen edge tests at BH 0.05 within contrast, for the reproduced test on the data and under the coverage-only null (one draw). The full-data brain run requires half of each group's samples and the half runs three samples, as in the evaluation.

Table S9 gives the calls. Under the null, which has no difference in PSI, the test makes 74% as many calls as on the data in the brain full data and about as many in GSE320507. The null rate follows the coverage ratio between the groups. In the GSE320507 full data, with r the ratio of the median LSV totals of the two groups, tests with $|\log_2 r| \le 0.5$, 0.5–1, 1–2, 2–3 and above 3 are called at rates 0.006, 0.18, 0.71, 0.86 and 0.93 under the null and 0.013, 0.20, 0.72, 0.86 and 0.94 on the data; in brain half 2 the rates are 0, 0.0003, 0.016, 0.17 and 0.44 under the null and

0.001, 0.003, 0.026, 0.19 and 0.44 on the data. Calls with a large pooled PSI difference are rarely reproduced, 20 of 750 with a difference above 0.2 in GSE320507 and 6 of 471 in brain.

In GSE320507, MAJIQ without removal replicates 991 genes on its comparator universe against 70 for Tealeaf. The coverage-only null removes 96% of its recomputed edges in each half, and 26 genes replicate after removal (Tealeaf 68). Restricting MAJIQ instead to tests with $|\log_2 r| \leq 1$ gives 15 replicated genes in both datasets (Tealeaf 190 in brain and 62 in GSE320507), and $|\log_2 r| \leq 0.5$ gives 8 (Tealeaf 143 and 41).

The count models of the other junction methods and of Tealeaf do not show this dependence. In GSE320507, junction counts redrawn within each LeafCutter cluster and each scQuint intron group from the proportions pooled over all pseudobulks, with each pseudobulk's observed total, give no BH discoveries for either method on the 171 cell-type pairs (on the data, 1,018 and 2,848). For Tealeaf, the count null of Section 4.9 on 200 random full-data tests per block structure (subject concentration 20, no row variation) rejects at 0.05 for 0 to 6% of converged internal and 2 to 4% of terminal tests in every class of the molecule-total ratio between the two cell types, including ratios above eightfold, where the MAJIQ null calls 93% of tests.

3 rMATS under the coverage-only null

Most rMATS calls in GSE320507 are mutually exclusive exons (83 to 87% of the BH calls in each half), and 96% of the called pairs have both exons in one annotated transcript, so the event compares junctions of different isoforms. With every event's inclusion count redrawn as in Section 4.12, pooled over all pseudobulks, PAIRADISE on four whole half-1 contrasts makes about as many BH calls of mutually exclusive exons as on the data (594 to 2,256 against 637 to 2,267) and more of skipped exons (40 to 180 against 17 to 79), and rejects 10 to 13% of skipped-exon and 40 to 51% of mutually-exclusive-exon tests at 0.05. Applied per test in the GSE320507 halves, the null removes 24 to 27% of the recomputed rMATS tests, at similar rates in every event type, against 0.4 to 0.8% for Tealeaf, LeafCutter and scQuint, whose replicated genes change by at most two (Tealeaf / comparator on the comparator universes, 67 / 126 for LeafCutter and 54 / 67 for scQuint). rMATS replicates 184 genes without removal and 133 after (Tealeaf 60 on both). In brain, rMATS replicates 4 genes with or without removal.

4 Results without the coverage-only filter

Without removal the split replication of Supplementary Table S1 is, for Tealeaf / comparator on the matched universe, 182 / 165 for LeafCutter, 206 / 27 for MAJIQ (2,764 genes) and 192 / 135 for scQuint, and on the comparator universe 182 / 169, 197 / 28 (3,003 genes) and 170 / 124; the SUPPA2, primer-aware SUPPA2 and rMATS columns are unchanged. Long-read agreement without removal is $A_{100} = 0.938$ for MAJIQ (85 of 100 agree, 150 of 200) and 0.914 for scQuint (90 of 100), with 1,565 and 8,497 evaluable features; Tealeaf (35,233 evaluable), LeafCutter and rMATS have the same A_{100}. On the 28 long-read pairs, Tealeaf has 3,663 discoveries in 3,045 gene–cell-type pairs, scQuint 2,234 pairs and MAJIQ 873; over all pairs, Tealeaf has 10,830 discoveries in 9,085 pairs, scQuint 8,835, MAJIQ 2,189 (on 170 pairs) and rMATS 570. The event classes of Table S2 and the calibration of Section 4.9 use the Tealeaf tests without removal.